FEMCI – The Book

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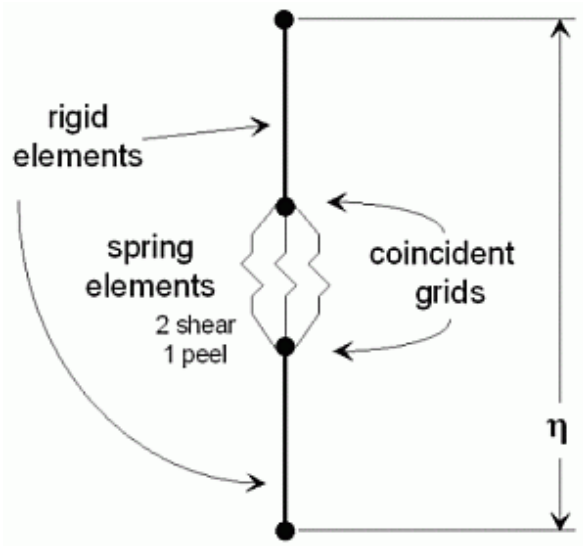
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Adhesive Modeling in Bonded Joints

A practical finite element approach has been developed to model the adhesive in a bonded joint. Numerical examples have shown good agreement with classical solutions.

The method uses a gap the thickness of the adhesive, two rigid elements, and three zero- length spring elements between coincident nodes. One rigid element stretches from one adherend to a node at the center of the gap, while the second rigid element stretches from the other adherend to a coincident node also at the center of the gap.



The spring elements connect the two rigid elements between the coincident nodes. The figure shows the layout. For clarity, the coincident nodes are shown separated. No rotational springs are used in this modeling technique.

Forces, stresses, strains, etc., can be recovered directly for the adherend elements. Recovery of spring forces and deformations can in turn be used to determine the stresses and strains in the adhesive.

The various spring stiffness values are determined using the following equations. Note that there are different values depending on whether the spring elements are internal to, along the edge of, or at a corner of the finite element mesh.

A_{el} = Element area
G = Shear modulus of the adhesive
η = Adhesive thickness
E = Young's modulus of the adhesive
v = Poisson's Ratio

Shear spring constant, *ks*, in both lateral directions,

$$k_{s-internal} = 2k_{s-edge} = 4k_{s-corner} = \frac{A_{el}G}{\eta}$$

$$k_{s-internal} = 2k_{s-edge} = 4k_{s-corner} = \frac{A_{el}G}{\eta}$$

Peel spring constant, k_p, through the thickness,

$$k_{p-internal} = \frac{A_{el}\left(E_a\right)_{eff-internal}}{\eta}$$

where

$$\left(E_a\right)_{eff-internal} = E = 2G(1+\nu)$$

$$k_{p-edge} = \frac{A_{el}\left(E_a\right)_{eff-edge}}{2\eta}$$

where

$$\left(E_a\right)_{eff-edge} = \frac{E(1-\nu)}{\left(1-\nu-2\nu^2\right)}$$

$$k_{p-corner} = \frac{A_{el}\left(E_a\right)_{eff-corner}}{4\eta}$$

where

$$\left(E_a\right)_{eff-corner} = \frac{E}{\left(1-\nu^2\right)}$$

Output results for the adhesive can be calculated using the following equations:

$$\tau_i = \frac{\sqrt{f_{x,i}^2 + f_{y,i}^2}}{A_{el}} \qquad \gamma_i = \frac{\sqrt{\delta_{x,i}^2 + \delta_{y,i}^2}}{\eta}$$

$$\sigma_i = \frac{f_{z,i}}{A_{el}} \qquad \varepsilon_i = \frac{\delta_{z,i}}{\eta}$$

where
i = Grid number
x, y = Shear directions

z = Peel direction

δ = Spring deflection

ε = Peel strain

f = Spring force

γ = Shear strain

σ = Peel stress

τ = Shear stress

Coefficient of Thermal Expansion

Dr. William Case
April 14, 1994

By definition, thermal strain at temperature ***T*** is the change in length of a member, due to a change in temperature, ***(T - Tref)***, divided by the original length ***l*** of that member. Denoting thermal strain at temperature ***T*** as e_T

$$e_T(T) = \frac{\Delta l_{thermal}}{l} \quad (1).$$

The thermal strain is zero at the reference temperature, T_{ref} .

Also, by definition, the coefficient of thermal expansion (CTE or α) at temperature ***T*** is defined as

$$\alpha(T) = \frac{d e_T}{dT} \quad (2).$$

Therefore,

$$e_T(T) = \int_{Tref}^{T} \alpha(t)dt \quad (3),$$

where ***t*** is a dummy variable of integration. Define ***α(T)*** to be the average value of alpha over the temperature range from T_{ref} to ***T***.

Then,

$$\overline{\alpha}(T) = \frac{\int_{Tref}^{T} \alpha(t)dt}{T - T_{ref}} \quad (4).$$

Then, the thermal strain can be written as:

$$e_T \equiv \frac{\Delta l_{thermal}}{l} = \bar{\alpha}\left(T - T_{ref}\right) \quad (5).$$

Generally, thermal expansion data is given as a curve of $\frac{\Delta l_{thermal}}{l}$ **vs.** ***T***.

Based on equations (2) and (5), it is seen that:

$$\alpha(T) = \frac{d}{dT}\left(\frac{\Delta l_{thermal}}{l}\right) \quad (6),$$

Whereas

$$\bar{\alpha}(T) = \frac{1}{T - T_{ref}}\left(\frac{\Delta l_{thermal}}{l}\right) \quad (7).$$

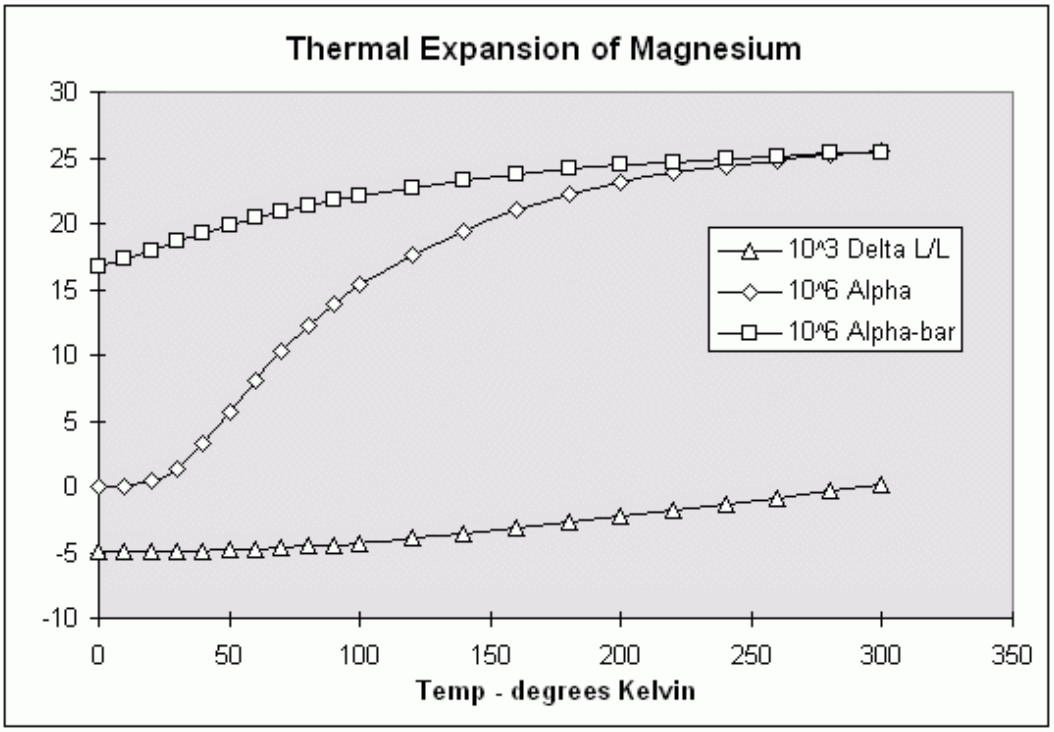
NASTRAN requires that $\bar{\alpha}$ be input, *not* α. However, the two are equal when α is constant.

An example of the calculation of $\bar{\alpha}$, along with the NASTRAN material data input, is given below.

Thermal Expansion of Magnesium

(Delta L/L and Alpha data from NBS Monogram 29: *Thermal Expansion of Technical Solids at Low Temperatures*)

Temp, deg K	10^3 Delta L/L	10^6 Alpha	10^6 Alpha-bar
0	-4.90	0.0	16.7
10	-4.90	0.1	17.3
20	-4.90	0.4	17.9
30	-4.89	1.4	18.6
40	-4.86	3.3	19.2
50	-4.83	5.7	19.9
60	-4.75	8.1	20.4
70	-4.66	10.3	20.9
80	-4.54	12.2	21.3
90	-4.41	13.9	21.7
100	-4.27	15.4	22.1
120	-3.93	17.6	22.7
140	-3.56	19.4	23.3
160	-3.16	21.0	23.8
180	-2.73	22.2	24.2
200	-2.27	23.2	24.4
220	-1.80	23.9	24.7
240	-1.32	24.4	24.9
260	-0.83	24.8	25.1
280	-0.33	25.2	25.3
300	0.18	25.5	25.4



NASTRAN Temperature Dependent Material Data Input for Magnesium
(using degrees Kelvin for temperature)

```
MAT1     11       6.5E6    2.4E6             .066     25.35E-6 293.
MATT1                                                  115
TABLEM1 115                                                                +T1151
+T1151   0.       16.7E-6 10.      17.3E-6 20       17.9E-6 30.      18.6E-6 +T1152
+T1152   40.      19.2E-6 50.      19.9E-6 60.      20.4E-6 70.      20.9E-6 +T1153
+T1153   80.      21.3E-6 90.      21.7E-6 100.     22.1E-6 120.     22.7E-6 +T1154
+T1154   140.     23.3E-6 160.     23.8E-6 180.     24.2E-6 200.     24.4E-6 +T1155
+T1155   220.     24.7E-6 240.     24.9E-6 260.     25.1E-6 280.     25.3E-6 +T1156
+T1156   300.     25.4E-6 ENDT
```

Dr. William Case
April 14, 1994

Craig Bampton Models

Scott Gordon
NASA Goddard Space Flight Center
May 6, 1999
(updated January 2008)

Background

Who is *Craig Bampton?*
Coupling of Substructures for Dynamic Analysis
by Roy R. **Craig** Jr. and Mervyn C. C. **Bampton**
AIAA Journal, Vol. 6, No. 7, July 1968

What is the Craig-Bampton Method?

- Method for reducing the size of a finite element model.
- Combines motion of boundary points with modes of the structure assuming the boundary points are held fixed.
- Similar to other reduction schemes

Guyan Reduction

$$\{U\} = [\phi]\{Ua\}$$

$$\text{Where } [\phi] = -[Koo]\text{-}1[Koa]$$

$$\{Ua\} = \text{A-set points}$$

Modal Decoupling

$$\{U\} = [\phi]\{q\}$$

$$\text{Where } [\phi] = \text{Mode Shapes}$$

$$\{q\} = \text{Modal dof's}$$

Craig-Bampton

$$\{U\} = [\phi]\{x_{cb}\}$$

Where $[\phi]$ = C-B Transformation

$\{x_{cb}\}$ = C-B Dof's = boundary + modes

Why is the C-B Method Used?

- Allows problem size to be reduced.
- Accounts for both mass and stiffness (unlike Guyan reduction)
- Problem size defined by frequency range.
- Allows for different boundary conditions at interface (unlike modal decoupling)

Craig-Bampton Theory

Equation of Motion (ignoring damping)

$$[M_{AA}]\{ii_A\} + [K_{AA}]\{u_A\} = \{F(t)\} \qquad (1)$$

The Craig-Bampton transform is defined as:

$$\{u_A\} = \left\{\begin{matrix} u_b \\ \hline u_L \end{matrix}\right\} = \begin{bmatrix} I & 0 \\ \phi_R & \phi_L \end{bmatrix} \begin{Bmatrix} u_b \\ q \end{Bmatrix} \qquad (2)$$

Where:

u_b = boundary DOFs

u_L = internal (leftover) DOFs

ϕ_R = rigid body vector

ϕ_L = fixed base mode shapes

q = modal DOFs

The 2X2 matrix in Equation 2 is the Craig-Bampton transformation Matrix.

Combining equations (1) and (2) and pre-multiplying by $[\phi_{cb}]^T$

$$\phi_{cb}^T [M_{AA}] \phi_{cb} \begin{Bmatrix} \ddot{u}_b \\ \ddot{q} \end{Bmatrix} + \phi_{cb}^T [K_{AA}] \phi_{cb} \begin{Bmatrix} u_b \\ q \end{Bmatrix} = \phi_{cb}^T \begin{Bmatrix} F_b \\ F_L \end{Bmatrix} \qquad (3)$$

Define the C-B mass and stiffness matrices as

$$[M_{cb}] = \phi_{cb}^T [M_{AA}] \phi_{cb} = \begin{bmatrix} M_{bb} & M_{bq} \\ M_{qb} & M_{qq} \end{bmatrix} \qquad (4)$$

$$[K_{cb}] = \phi_{cb}^T [K_{AA}] \phi_{cb} = \begin{bmatrix} K_{bb} & 0 \\ 0 & K_{qq} \end{bmatrix} \qquad (5)$$

Write equation (3) using equations (4) and (5)

$$\begin{bmatrix} M_{bb} & M_{bq} \\ M_{qb} & M_{qq} \end{bmatrix} \begin{Bmatrix} \ddot{u}_b \\ \ddot{q} \end{Bmatrix} + \begin{bmatrix} K_{bb} & 0 \\ 0 & K_{qq} \end{bmatrix} \begin{Bmatrix} u_b \\ q \end{Bmatrix} = \begin{Bmatrix} F_b \\ 0 \end{Bmatrix} \qquad (6)$$

Where input forces are applied at the boundary only (F=0)

Important properties of the C-B mass and stiffness matrices:

- M_{bb} = Bounday mass matrix -- total mass properties translated to the boundary points

$$\left[M^{cg}\right] = \phi_{RB}^{cg^T} \left[M_{bb}\right] \phi_{RB}^{cg} \qquad (7)$$

- K_{bb} = Interface stiffness matrix -- stiffness associated with displacing one boundary DOF while other are held fixed.
    - If the boundary point is a single gird (i.e. non-redundant) then $K_{bb} = 0$

- If the mode shapes have been mass normalized (typically they are) then

$$K_{qq} = \begin{bmatrix} \ddots & & 0 \\ & \lambda & \\ 0 & & \ddots \end{bmatrix} \qquad \lambda_i = k_i / m_i = \omega_i^2$$

$$M_{qq} = \begin{bmatrix} \ddots & & 0 \\ & I & \\ 0 & & \ddots \end{bmatrix} \qquad (8)$$

We can finally write the dynamic equation of motion (including damping) using the C-B transform as

$$\begin{bmatrix} M_{bb} & M_{bq} \\ M_{qb} & I \end{bmatrix} \begin{Bmatrix} \ddot{u}_b \\ \ddot{q} \end{Bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 2\zeta\omega \end{bmatrix} \begin{Bmatrix} \dot{u}_b \\ \dot{q} \end{Bmatrix} + \begin{bmatrix} K_{bb} & 0 \\ 0 & \omega^2 \end{bmatrix} \begin{Bmatrix} u_b \\ q \end{Bmatrix} = \begin{Bmatrix} F_b \\ 0 \end{Bmatrix} \quad (9)$$

Where,

$2\zeta\omega$ = modal damping

ζ = % critical damping

Summary of C-B Theory

- C-B Mass and Stiffness Matrices fully define system
  Dynamics problem solved using C-B DOFs
- C-B boundary DOFs provide location to apply BC's and Forces or to couple with another structure
- C-B transform is used to calculate physical responses from C-B responses.

How to Create a C-B Model

```
$ C-B Output File
ASSIGN,OUTPUT4='cb_ex1.kmnp',STATUS=unknown,UNIT=31
ID GORDON,CB_EX1
$ Normal Modes Solution
SOL SEMODES
TIME 5
DIAG 8,14
$ C-B DMAP
INCLUDE 'path/cb103_2005r2b.dmp'
CEND
$
TITLE = Craig-Bampton Example 1
ECHO = NONE
METHOD = 1
$ SPC = 1
$ Request modal effective mass output
MEFFMASS(ALL) = YES
$
$ Request Output Grid Point Forces, Element Forces, and Element Stresses
$ This is recovered in the .F06 file
SET 100 = 209,210
SET 101 = 110,114
GPFORCE = 100
ELFORCE = 101
ELSTRESS = 101
$
BEGIN BULK
$
PARAM,AUTOSPC,YES
PARAM,GRDPNT,0
PARAM,WTMASS,0.00259
$
$ Print G-set (U1 in the DMAP) & R-set internal order
PARAM,USETSEL,0
PARAM,USETSTR1,U1
PARAM,USETSTR2,R
PARAM,USETPRT,0
$
$ Acceleration Output
$ Recover accleration response at GRID 11
$ This is output in the .kmnp file and processed in FLAME
USET1   U1      123456 11
$
$ Define number of modes to recover (or frequency range)
EIGRL   1       10      0       MASS
$
$ Boundary (R-set) Defined on SUPORT cards
$ 1 boundary point x 6 DOFs
$ = 6 physical boundary points
SUPORT 999      123456
SPC     1       999 123456      0.
$
$ Don't forget the rest of your bulk data
INCLUDE 'bulkdata.bdf'
$ ENDDATA
```

What is created?

- File cb_ex1.kmnp which contains C-B stiffness and mass matrices (k,m), net CG ltm (n), and the C-B transformation matrix (phi_g)
- The .kmnp file is in NASTRAN binary output4 format
- K and M size is [C-B DOFs (boundary + modal) x C-B DOFS]
- phi_g size is [G-set rows x C-B DOFs]
- Net CG LTM recovers CG accelerations and I/F Forces, Size is [6+boundary DOFs x C-B DOFs]

How do you use this?

- Solve dynamics problem for C-B DOF response using the K and M matrices
- Transform C-B responses using phig to get physical responses

Load Transformation Matrices (LTMs)

- LTM is a generic term referring to the matrix used to transform from C-B DOFs to physical DOFs (also referred to at OTMs, ATMs, DTMs …)
- In its simplest form, the LTM is simply the phi$_g$ matrix

$$\{\ddot{U}_G\} = [\phi_{cb}] \begin{Bmatrix} \ddot{u}_b \\ \ddot{q} \end{Bmatrix} \qquad (10)$$

(Only the rows corresponding to the physical DOFs of interest are needed)

There are other useful LTMs that can be created:

- Interface (I/F) forces
- Net CG accelerations
- Stress and force LTMs

I/F Force LTM (created by C-B DMAP)

IF Force =

$$\begin{bmatrix} M_{bb} & M_{bq} & K_{bb} \end{bmatrix} \begin{Bmatrix} \ddot{u}_b \\ q \\ u_b \end{Bmatrix} \qquad (11)$$

(If boundary is non-redundant, then Kbb = 0)

Net CG LTM (created by C-B DMAP)

Net CG acceleration =

$$\left(\left[\phi_{rb}^{CG}\right]^T\left[M_{bb}\right]\left[\phi_{rb}^{CG}\right]\right)^{-1}\left[\phi_{rb}^{CG}\right]^T\left[M_{bb}\ M_{bq}\ K_{bb}\right]\begin{Bmatrix}\ddot{x}_b\\ \ddot{q}\\ x_b\end{Bmatrix} \qquad (12)$$

where

$$\left(\left[\phi_{rb}^{CG}\right]^T\left[M_{bb}\right]\left[\phi_{rb}^{CG}\right]\right)$$

is the mass matrix about the CG (6 x 6)

and

$$\left[\phi_{rb}^{CG}\right]$$

is the rigid body transform from I/F to CG (BDOF x 6)

PHIZ LTM

Allows physical displacements to be calculated from C-B accelerations:

$$\{X_G\} = [PHIZ]\begin{Bmatrix}\ddot{x}_b\\ \ddot{q}\end{Bmatrix} \qquad (13)$$

- Same as modal acceleration approach in NASTRAN
- Useful in calculating relative displacements between DOFs
- Also used to calculate stresses and forces which are a function of displacements
- Calculated from C-B dmap using **PARAM,PHZOUT,1**

LTM's can be created using FLAME, MATLAB or using DMAP

LTM's can (and usually do) contain multiple types of responses

$$LTM = \begin{bmatrix} [Net\,CG] \\ [I\ /\ F\ Force] \\ [Accel] \\ \begin{bmatrix} Element \\ Forces \end{bmatrix} \end{bmatrix} \qquad (14)$$

LTM's can be used to recover responses for nested C-B models

$$\{X^{cb1}\} = [\phi_{cb1}][\phi_{cb0}^{cb1}] \begin{Bmatrix} \ddot{x}_b^{cb0} \\ \ddot{q}^{cb0} \end{Bmatrix} \qquad (15)$$

Where

$$\phi_{cb0}^{cb1}$$

Is the row partition of CB_1 DOFs from the CB_0 PHI_G matrix.

Checking C-B Models and LTMs

C-B Models and LTMs should be verified to make sure that they have been created correctly (especially for complicated LTM's or nested C-B models)

C-B Mass and stiffness matrices can be checked by computing free-free and fixed-base modes C-B boundary Mass matrix can be transformed to CG and compared with NASTRAN GPWG

LTMS can be checked by applying unit acceleration at the boundary:

$$\{X_{resp}\} = \left[\begin{array}{c|c} b & q \end{array}\right]^{LTM} \begin{Bmatrix} \phi_{RB}^{b} \\ 0 \end{Bmatrix} \qquad (16)$$

Where

ϕ_{RB}^{b}

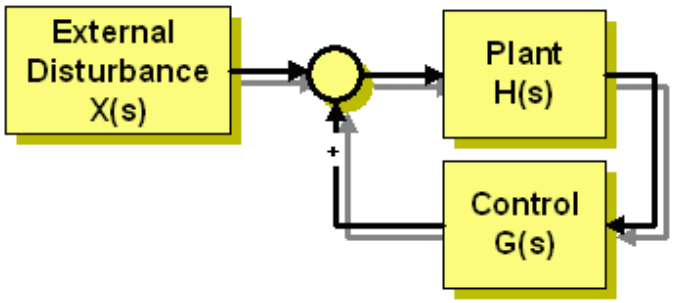
Is the boundary rigid body vector (b x 6)

- Each response column represents acceleration in a single direction
- Accelerations should be in correct directions
- Forces should recover weight or correct moments
- Unit acceleration applied to PHIZ can be checked by gravity run with physical model and comparing displacements

Development of Fixed Base Shake Transfer Functions

Timothy Carnahan
Goddard Space Flight Center
January 23, 1996

The block diagram below is for a typical feedback servo control system in an instrument.



The plant, H(s), will generate some error signal of a sensor. In motor systems this is typically an error between the rotor and the stator. The controller senses these errors and applies a correction load. For rigid systems the external load would be applied to the mass of the system to generate a disturbance acceleration. The more complex systems are flexible. This could mean that the flexibility is in a rotational shaft or between the point of application of an external load and the instrument. For the CIRS scan mechanism a flexible plant model was desired to couple spacecraft disturbances at the base of the instrument with sensor errors between the reference interferometer and the scan mirror. Since the S/C is supplying a disturbance without concern for the response of the instrument, the plant model assumes the instrument is fixed to ground. Further, the S/C generates disturbance forces and moments at the base of the instrument.

The following is the methodology and derivations developed to provide the controls analyst with H(s). This is expressed as a transfer function of coefficients to a numerator/denominator pair.

The general equation of motion that governs an undamped systems without external forces is:

$$M\ddot{u}_B + Ku = 0 \quad (1)$$

For a system with fixed base motions acceleration, $\ddot{u}_B$

Equation (1) becomes:

$$M(\ddot{u} - \ddot{u}_B) + K(u - u_B) = -M\ddot{u}_B \quad (2)$$

If we define $u_r = u - u_B$ and $F_B = -M\ddot{u}_B$

Equation (2) becomes:

$$M\ddot{u}_r + Ku_r = F_B \quad (3)$$

Expressing u_r in model coordinates we get

$$u_r = \sum_i \phi_i q_i \quad (4)$$

Where ϕ_i is a fixed base mode shape for mode i and q_i are the model coordinates.

Converting equation (3) to model coordinates and pre-multiplying by ϕ^T

$$\phi^T M \phi \ddot{q} + \phi^T K \phi q = \phi^T F_B \quad (5)$$

For the modal mass, m = $\phi^T M \phi$

The modal stiffness, k = $\phi^T K \phi$

And including generalized modal damping of $2\xi m$

We get:

$$m\ddot{q} + 2\xi m \dot{q} + kq = \phi^T F_B \quad (6)$$

Solving equation (6) for q(s) and using $\omega^2 = \frac{k}{m}$

$$q(s) = \frac{\frac{\phi^T F_B}{m\omega^2}}{\frac{1}{\omega^2}s^2 + \frac{2\xi}{\omega}s + 1} \quad (7)$$

Using equation (7) in equation (4) to solve for u_r

$$u_r = \sum_i \frac{\phi^T\left(\frac{AO}{m\omega^2}\right)}{\frac{1}{\omega^2}s^2 + \frac{2\xi}{\omega}s + 1} \quad (8)$$

Solving for u

$$u_B + u_r = u_B + \sum_i \frac{\phi_c\left(\frac{AO}{m\omega^2}\right)}{\frac{1}{\omega^2}s^2 + \frac{2\xi}{\omega}s + 1}$$

For relative rotor/stator T.F. we want the difference between the two or, ustator - urotor = durotor-stator which expands to

$$\Delta u_{rotor-stator} = u_B + u_{r_{rotor}} - (u_B + u_{r_{stator}}) = u_{r_{rotor}} - u_{r_{stator}} \quad (10)$$

$$\Delta u_{rotor-stator} = \sum_i \frac{\phi_{rotor}\left(\frac{AO}{m\omega^2}\right)}{\frac{1}{\omega^2}s^2 + \frac{2\xi}{\omega}s + 1} + \sum_i \frac{\phi_{stator}\left(\frac{AO}{m\omega^2}\right)}{\frac{1}{\omega^2}s^2 + \frac{2\xi}{\omega}s + 1}$$

$$\Delta u_{rotor-stator} = \sum_i \frac{(\phi_{rotor} - \phi_{stator})\left(\frac{AO}{m\omega^2}\right)}{\frac{1}{\omega^2}s^2 + \frac{2\xi}{\omega}s + 1} \quad (11)$$

For absolute stator T.F. we want a complete solution to u. Remember that that equations of motion, (2), was in terms of acceleration and therefore

$$u_B(s) = \frac{\ddot{u}_B}{s^2}$$

$$u_{stator} = \frac{\ddot{u}_B}{s^2} + \sum_i \frac{\phi_{stator}\left(\frac{AO}{m\omega^2}\right)}{\frac{1}{\omega^2}s^2 + \frac{2\xi}{\omega}s + 1} \quad (12)$$

Equations (11) and (12) needs the inputs:

m - the modal mass for mode i

ϕ_i - the mode i cantilevered shapes.

ω_i - the eigenvalue for mode i

ξ_i - generalized modal damping

and AO.

For mass normalized eigenvectors m is unity and can be eliminated from the problem.

AO is defined as $-\phi^T M \ddot{u}_B$

The 'motion' program processes the output of a normal modes run, rigid format 3, and generate the H(s) T.F. for Matlab. The DMAP for UAI NASTRAN version 11.8 to solve for AO and output to an OUTPUT4 file is shown below.

```
ASSIGN AOout=aOout.ot4,NEW,USE=OUTPUT4,TYPE=FORMATTED
ALTER 188 $
MATGEN BGPDT,EQEXIN,CSTM/UB/102/0/ $ GENERATE Ub
MPYAD MGG,UB /FB/0/ -1/0/ $
MPYAD PHIG,FB, /AO/-l/l/O/ $
OUTPUT4 A0,,,,//*AOout*/0 /*FULL*/ $
ENDALTER $
```

The eigenvalues and eigenvectors are read from a punch file. In order for the post-processing program to work the following guidelines must be observed:

1. Because m is not exported to a file the eigenvectors must be mass normalized. (Therefore, m is the density matrix.)

2. Do not use DYNRED, as it changes the set definitions and the DMAPS MPYAD does not use the correct size arrays.

3. Use DISP(PUNCH)=set id to output the grid points of interest. The punch files are much smaller when only a set, the rotor/stator gid and the input gid, are in the punch file. The punch file includes the eigenvalues.

The 'motion' program is written in "C" to read the NASTRAN data and convert it into MATLAB m-files in the form of MATLAB numerator and denominator transfer functions. This program is located in /home/tim/bin and was written for SGI machines. Motion has been ported to an IBM RS6000. The source code is in the directory /home/tim/bin/SRC/NASTRAN. The data used by the "motion" program will come from a punch file which contains the eigenvectors and eigenvalues and the OUTPUT4 file of the A0 matrix. With these the program asks for the grid points and dof for the rotor and the stator and the direction of input disturbance. When the desired output is to know absolute position, stator only, the same grid id is entered for rotorr and the slator. Motion will add the needed MATl,AB commands to generate both the position and acceleration transfer functions for absolute displacement. The MATLAB variable names of these absolute transfer functions is pos_num and pos_den for position transfer functions and acc_num and acc_den for acceleration transfer functions. In the case of relative transfer functions only the position transfer functions are generated and the MATLAB variable names are plant_num and plant_den. The translational units of the transfer function are in whatever units the NASTRAN model was in. The rotational units are in degrees.

The 'motion' program is coded with the following maximums: Number of eigenvalues = 230. Note: matlab can not accurately handle such a large number of eigenvectors/T.F.'s. A method of mode selection is needed to provide only significant modes for controls work. The generalized modal damping, ~, is set to 0.1% of critical. This damping values depend on a number of mechanical characteristics from the magnitude of the applied structure to the type of assembly. For typical micro- gravity disturbances on small, < 100 lbs, a value of 0.1% is considered conservative.

Tim Carnahan

Eigenvalue Extraction Methods in NASTRAN

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November 29, 1995

Introduction

This document will discuss 3 of the commonly used eigenvalue extraction techniques used in NASTRAN: Modified Givens, Inverse, Lanczos. In addition, 2 methods of automated model dynamic reduction will also be investigated: AUTOOMIT(static condensation or Guyan Reduction) and Generalized Dynamic Reduction.

Since it's beyond the scope of this report (and my scope!) to discuss the theories of the various eigenvalue extraction techniques, this report will stick mainly to a general discription of the technique, the pluses and minuses, and other considerations to be aware of. Following the detailed descriptions of these methods will be a summary table.

Methods

This section will discuss three of the NASTRAN methods used to extract eigenvalues for a dynamics analysis. These methods can be used alone, or in conjunction with dynamic reduction techniques.

Modified Givens: (MGIV) This method is, for most problem sizes, the slowest method of eigenvalue extraction. MGIV will solve for ALL eigenvalues in the NASTRAN model (eigenvectors will only be calculated for the range of frequencies specified). As a result, this method is most useful if all the eigenvalues are of equal interest. For most problems, MGIV is only practical if the model is small, or if an ASET or OMIT set is provided in the model in order to decrease the number of dynamic degrees of freedom (DOF). When using MGIV, NASTRAN will, by default, use AUTOOMIT to remove all massless degrees of freedom from the ASET DOF in order to compact the matrices.

As stated earlier, for practical purposes, MGIV requires that the model be reduced in size. This can be done by either specifying which degrees of freedom to keep in the model (ASET), or which degrees to remove from the model (OMIT). The problem with reducing the model is that it introduces error into the results, particularly for higher modes, as not all the structural mass and stiffness is taken into consideration. This can result in the shifting of mode frequencies as well as missing frequencies completely. Deciding what degrees of freedom to keep is an art unto itself. UAI/NASTRAN has automated the process somewhat with the use of AUTOOMIT.

Inverse: (SINV) This method is faster than MGIV for eigenvalue extraction, but still uses the entire mass and stiffness matrices of the model. SINV is very efficient for extracting a specific range of frequencies. Unlike MGIV, SINV only solves for the eigenvalues in the frequency range of interest. Also unlike MGIV, SINV does not need to remove massless DOF's in order to run efficiently. In fact, SINV takes advantage of the sparseness of the matrices caused by massless degrees of freedom. As a result, using

AUTOOMIT to remove massless degrees of freedom from an SINV run will actually Increase model run time. Since SINV uses all the DOF's in the model and does not require ASET's for reasonable run times the solution accuracy is very good and, for similar run times, is more accurate than an MGIV model run with an ASET.

Lanczos: (LANCZ) The Lanczos method is, for most applications, the fastest, accurate solution method in NASTRAN. LANCZ is faster than both MGIV and SINV and is just as accurate for the models tested. LANCZ is also a sparse matrix solver and, like SINV, will run slower if the matrices are compacted by removing massless DOF. This method of extraction seems the best choice for most NASTRAN models. Even with compacted matrices, LANCZ can be faster than any of the other methods mentioned. However, if you have large, dense matrices (i.e.. very few massless DOF), using the MGIV method along with Generalized Dynamic Reduction may be the fastest method.

Dynamic Reduction Techniques

Dynamic reduction is a way of reducing the number of dynamic degrees of freedom in a finite element model and therefore reducing the size of the problem to be solved. This is particularly important when using MGIV as its solution time is dramatically decreased when the ASET can be reduced. One consequence of dynamic reduction, however, is the introduction of error. Since dynamic reduction removes degrees of freedom from the ASET it can cause mode frequencies to shift, and also may miss modes altogether.

AUTOOMIT: AUTOOMIT is a tool for automatically performing dynamic reduction on a model. AUTOOMIT is a case control command that creates an OSET (OMIT cards) automatically, given various parameters. It will automatically remove all massless DOF when using the MGIV method. AUTOOMIT can also be used to remove other DOF's from the ASET. AUTOOMIT eliminates DOF based on their mass to stiffness ratio. A high ratio means a DOF with high mass and relatively low stiffness. In general, this is more likely to be a DOF that will contribute to a lower frequency mode than a DOF that has low mass and high stiffness. AUTOOMIT can be requested to omit DOF's based on a value for the mass to stiffness ratio ("AUTOOMIT(EPS=x) =YES"), based on DOF mass ("AUTOOMIT(MASS=y) =YES"), or a certain percent of DOF's can be requested to be retained ("AUTOOMIT(KEEP=z) =YES").

AUTOOMIT works best in models with good, discreet mass distribution (i.e. - use of concentrated masses). For a model with a more homogeneous mass and stiffness distribution, AUTOOMIT will not produce an ASET with points distributed nicely around the entire model. This can increase errors as some more localized modes may be missed. In general, AUTOOMIT is a good tool to start generating an ASET for a model. In order to improve accuracy of the results, additional DOF's may have to be added later.

Generalized Dynamic Reduction: (DYNRED) Generalized Dynamic reduction (as the name implies) is not a method of eigenvalue extraction, but rather a method to reduce the size of the model other than using ASETs or OMITs. It also produces results much faster than using static condensation. In essence, the way that DYNRED works is by approximating the mode shapes of the model first, and then solving for the actual modes. It generates scalar points, based on the number of modes desired, and uses those scalar points to approximate mode shape. These represent the generalized coordinates for the mode shape approximation. In UAI NASTRAN they are created automatically.

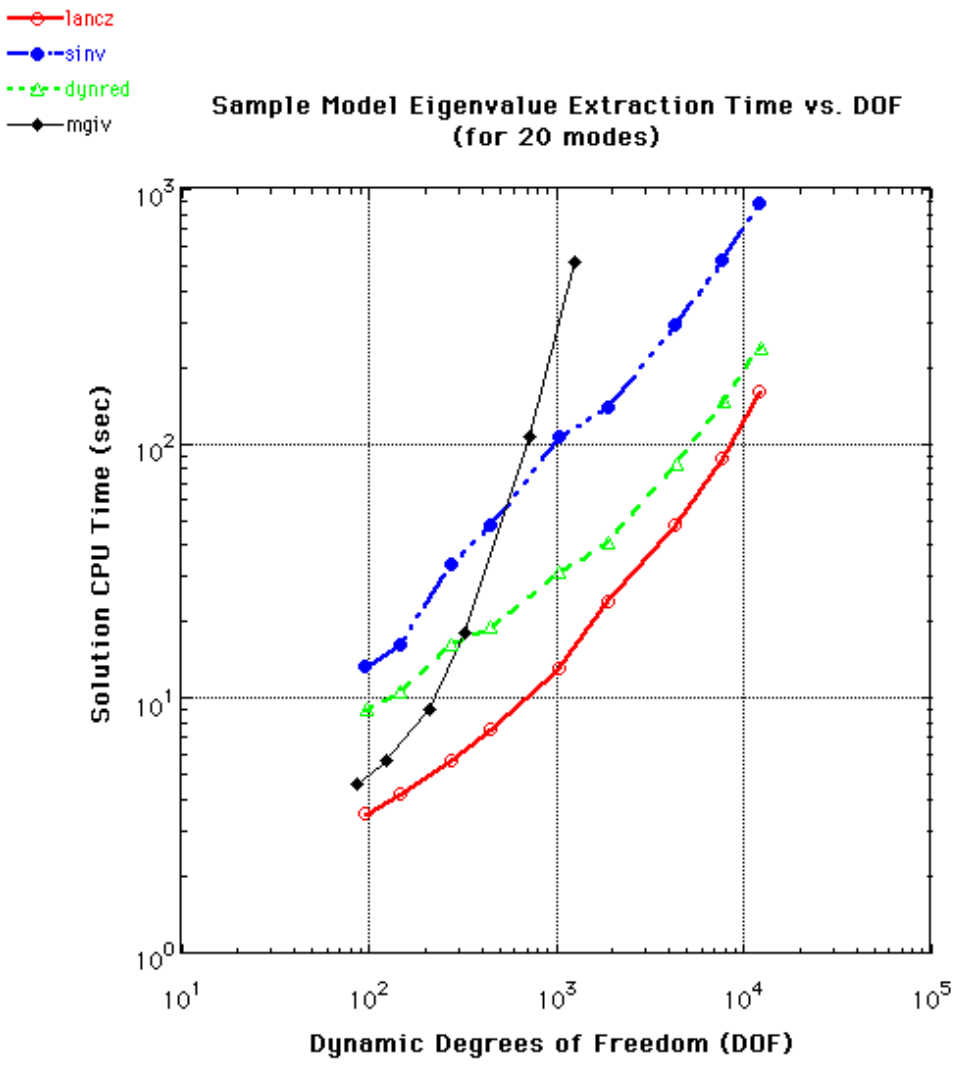
DYNRED, used in conjunction with SINV or MGIV, is generally a fast and accurate aid in extracting eigenvalues. For a sparse matrix LANCZ is still faster, but DYNRED is still no slouch. Since it approximates mode shape to begin with, it does not generate answers as accurately as SINV or LANCZ, or MGIV by itself. Generally, the first modes are very accurate, with accuracy tailing off toward the higher frequencies. Even at the higher modes, the accuracies are still rather good, especially when compared to MGIV with ASET or OMIT cards.

In UAI NASTRAN, using DYNRED requires a DYNRED card in the case control to specify the DYNRED ID, and a DYNRED card in the bulk data deck to specify the frequency range of interest. DYNRED still requires a METHOD card in the case control, and an EIGR card.

Analysis Results

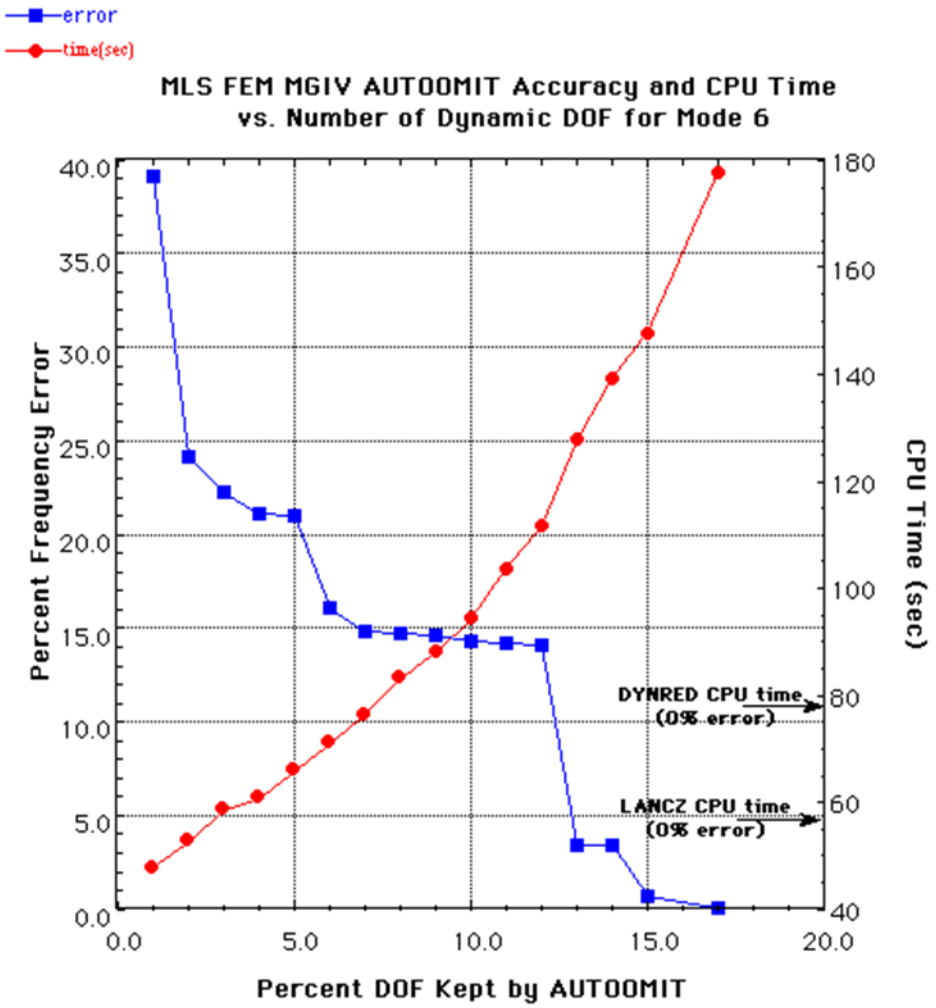
To better understand and compare the various eigenvalue extraction methods, several NASTRAN runs were made with a simple plate model. The model was a square plate held along one edge and held at one of the opposite corners. The model mesh started as a 5x5 mesh, but meshes of 6x6, 8x8, 10x10, 15x15, 20x20, 30x30, and 50x50 were also created from the same geometry. All these models were run using the previously mentioned eigenvalue extraction methods to calculate the first 20 modes. The results were then compared. All runs were made in UAI NASTRAN v8 and run on an IBM RS6000-410 computer.

Two specific pieces of data were compared: solution CPU time, and solution accuracy. For the sake of these comparisons, the SINV results were the baseline, as this method used the entire stiffness and mass matrices. All the methods, except for the MGIV with DYNRED runs, produced exactly the same modal frequencies. The error in the runs with DYNRED occurred only in the higher modes, and those errors were negligible. The main differences among the runs were the CPU times. The following table compares the times for the various methods based on the number of DOF in the model ASET.



As the problem size increases, the SINV and LANCZ methods begin to lose their appeal. The DYNRED method times are, essentially, unchanged.

Among dynamic reduction techniques, DYNRED generalized dynamic reduction is much more time efficient, and generally more accurate. The graph below compares AUTOOMIT static condensation to DYNRED for various percentages of DOF retained by AUTOOMIT. This chart is for the 6th mode of the MLS NASTRAN model. The percentage of frequency error for a given percentage of DOF kept by AUTOOMIT is shown in one line, with the CPU time shown on the other. The DYNRED solution time (which had no frequency error) is shown on the CPU time axis. For comparison, the Lanczos time is also shown:


error
time(sec)
MLS FEM MGIV AUTOOMIT Accuracy and CPU Time
vs. Number of Dynamic DOF for Mode 6
Percent Frequency Error
CPU Time (sec)
Percent DOF Kept by AUTOOMIT
DYNRED CPU time
(0% error)
LANCZ CPU time
(0% error)
0.0
5.0
10.0
15.0
20.0
25.0
30.0
35.0
40.0
40
60
80
100
120
140
160
180

Conclusion

For most model sizes and applications, the Lanczos eigenvalue extraction method is the most efficient. For smaller models, or models where all the eigenvalues are to be extracted, the MGIV method may be more useful. In addition, for models that do not have sparse matrices, the LANCZ and SINV methods may actually be slower than using other methods. For those cases, the MGIV method with DYNRED may be the best choice.

In addition to the example problems elaborated above, several other, more detailed NASTRAN models were run using the various extraction methods. These were models that had been used for real-world structural analysis. Although there have been some concerns that some solution extraction techniques may miss mechanism or duplicate modes, this was not seen in any of the problems run.

The DYNRED method is also a more acurate and faster aid to model dynamic reduction than using standard static condensation with AUTOOMIT. However, in cases where it's necessary to develop an ASET, or OSET for a model, such as when deciding where to place accelerometers for a vibration test, AUTOOMIT can be a very useful tool. It is, however, most useful for models with discrete, lumped masses and stiffness differences. For more homogeneous mass and stiffness distributions, it is not as efficient at selecting an ASET.

The following table summarizes the eigenvalue extraction methods detailed above:

	Performance Time vs Model Size				
Method	**small**	**med.**	**large**	**Accuracy**	**Best Applications**
MGIV	good	fair	poor	excellent	small models or when all eigenvalues are desired
SINV	good	good (1)	fair (2)	excellent	small to moderate size models when a range of frequencies are desired, and matrices are sparse
LANCZ	good	excellent (3)	excellent (3)	excellent	moderate to large models with sparse matrices
MGIV w/DYNRED	good	good	good	good	moderate to large models, particulary with dense matrices

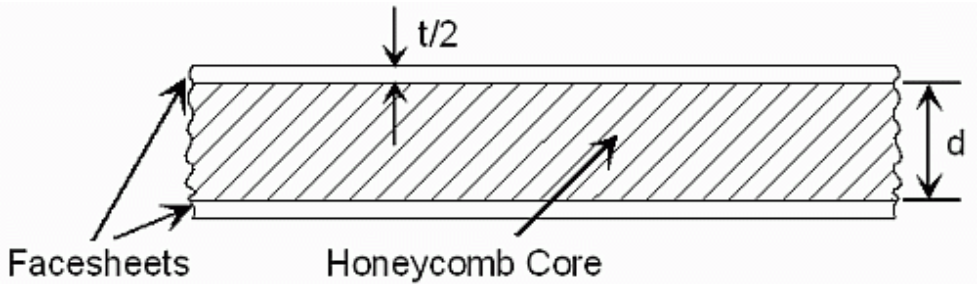
(1) For dense matrices, performance goes to fair

(2) For dense matrices, performance goes to poor

(3) For dense matrices, performance goes to good

Honeycomb Plates

Properties



Define the following terms as

t = total face sheet thickness (both sheets together)
d = total core depth
E = Young's modulus of face sheets
nu = Poissons ratio (Greek letter)
g = material density of face sheet material
gc = material density of core
I' = bending mom. of inertia/width of plate
K = membrane stiffness of plate (composite)
D = bending stiffness of plate (composite)
w = weight per unit area of plate (composite)

From classical plate theory, considering only the face sheets to carry in-plane stresses:

$$K = \frac{Et}{1-\upsilon^2}$$ membrane stiffness of plate **(Eq. 1)**

$$D = \frac{EI'}{1-\upsilon^2}$$ bending stiffness of plate **(Eq. 2)**

(NOTE: Shear flexibility of the plate is not included.)

Where I' is the bending moment of inertia of the honeycomb plate per unit width:

$$I' = \frac{2}{3}\left(\frac{d}{2} + \frac{t}{2}\right)^3 - \frac{d^3}{12}$$

For d >> t, **I'** simplifies to:

$$I' \approx \frac{td^2}{4} \quad \text{(Eq. 3)}$$

$$\therefore D = \frac{Et^3}{12\left(1-\upsilon^2\right)} \times 3\left(\frac{d}{t}\right)^2$$

The weight per unit area of the plate is:

$$w = gt + g_c d = gt\left[1 + \frac{g_c}{g}\left(\frac{d}{t}\right)\right]$$

Thus, the stiffness to weight ratios (membrane and bending) are:

$$K/w = \frac{E}{g\left(1-\upsilon^2\right)}\left[\frac{1}{1+\frac{g_c}{g}\left(\frac{d}{t}\right)}\right]$$

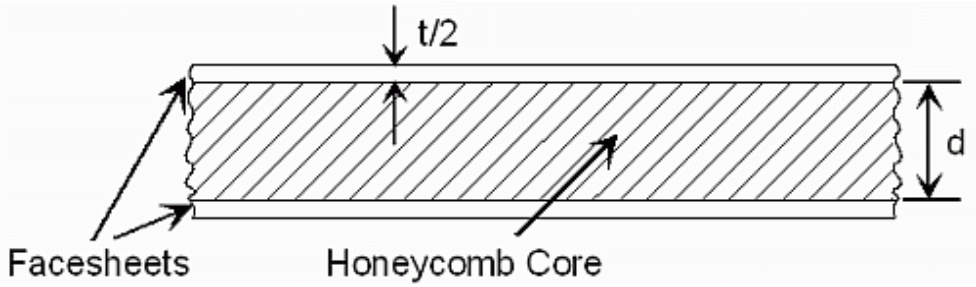
$$D/w = \frac{E}{g\left(1-\upsilon^2\right)}\left[\frac{t}{4}\,\frac{\left(\frac{d}{t}\right)^2}{1+\frac{g_c}{g}\left(\frac{d}{t}\right)}\right]$$

Define nondimensional stiffness/weight ratios:

$$k_1 = \frac{\left[\frac{K/E}{\left(1-\upsilon^2\right)}\right]}{\left[w/g\right]} = \frac{1}{1+\frac{g_c}{g}\left(\frac{d}{t}\right)}$$ Membrane

$$k_2 = \frac{\left[\frac{D/Et^2}{12\left(1-\upsilon^2\right)}\right]}{\left[w/g\right]} = \frac{\left[3\left(d/t\right)^2\right]}{\left[1+\frac{g_c}{g}\left(\frac{d}{t}\right)\right]}$$ Bending

PSHELL Card



Information needed:

T - Total facesheet thickness - this should include both facesheets (the value "t" in the figure above)
D - Total core thickness (the value "d" in the figure above)
Facesheet - Material properties
Honeycomb - Material properties, cell size, and gage

Format of the PSHELL Card:

1	2	3	4	5	6	7	8	9	10
PSHELL	PID	MID1	T	MID2	$12I/T^3$	MID3	T_S/T	NSM	

PSHELL and PID are self explanatory.

MID1, T

These terms are for in plane load of a honeycomb panel. Assume that the facesheets take all the load. Therefore:

> **MID1** references the material for the facesheets.
>
> **T** is the total thickness of the facesheets.

MID2, 12I/T3

These terms are for the bending load of a honeycomb panel. Assume the facesheets take all the load. Therefore:

> **MID2** references the material for the facesheets.
>
> **$12I/T^3$** is the inertia of a rectangular plate
>
> **I** for Honeycomb is approximately **$TD^2/4$** for thin facesheets and a thick core.
>
> If the facesheets are thicker relative to the core, then **I** is $[2/3][(D/2)+(T/2)]^3-[D^3/12]$ T is the total thickness of the facesheets.
>
> **D** is the thickness of the honeycomb core.

MID3, TS/T

These terms are for the shear load of a honeycomb panel. Assume that the honeycomb core takes all the load. Therefore:

> **MID3** references the material for the honeycomb core. The shear modulus and the poisson's ratio is required on the MATERIAL card for MID3 and is obtained from the HEXEL manuals. You need to know the cell size and gage, or the desired nominal density.
>
> $\mathbf{T_S}$ is the shear thickness, or for the case of honeycomb, it is the core thickness D.
>
> **T** is the total thickness of the facesheets.

NSM

The non structural mass is required for honeycomb panels. The density (rho) is ignored on the MID3 card, however, the mass of the honeycomb core and bonding material must be included. The NSM, which is given in mass per unit area, is determined by:

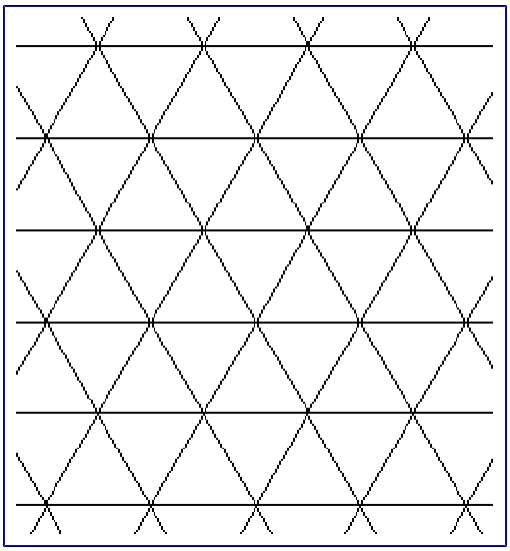
> **NSM = $(\mathbf{rho_{nominal}})(D)$,**
>
> where D is the thickness and $rho_{nominal}$ is often given in **pcf**, pounds per cubic foot. Be careful of your units when calculating NSM!

Jim Loughlin

Isogrid Plates

An Isogrid panel is a plate (or face sheet) with triangular integral stiffening ribs (often called stringers). The triangular pattern was found to be very efficient because it takes advantage of the fact that triangular trusses are very efficient structures. The term isogrid is used because the structure acts like an isotropic material.

The first figure shows the rib pattern for an isogrid plate. While other types of rib patterns can also be designed, e.g., square or rectangular, the triangular pattern is the only true isogrid.



The figure below shows the top view of a single isogrid cell. The flange is shown as a hidden line through the face sheet. The terms shown are defined as:

b - rib thickness

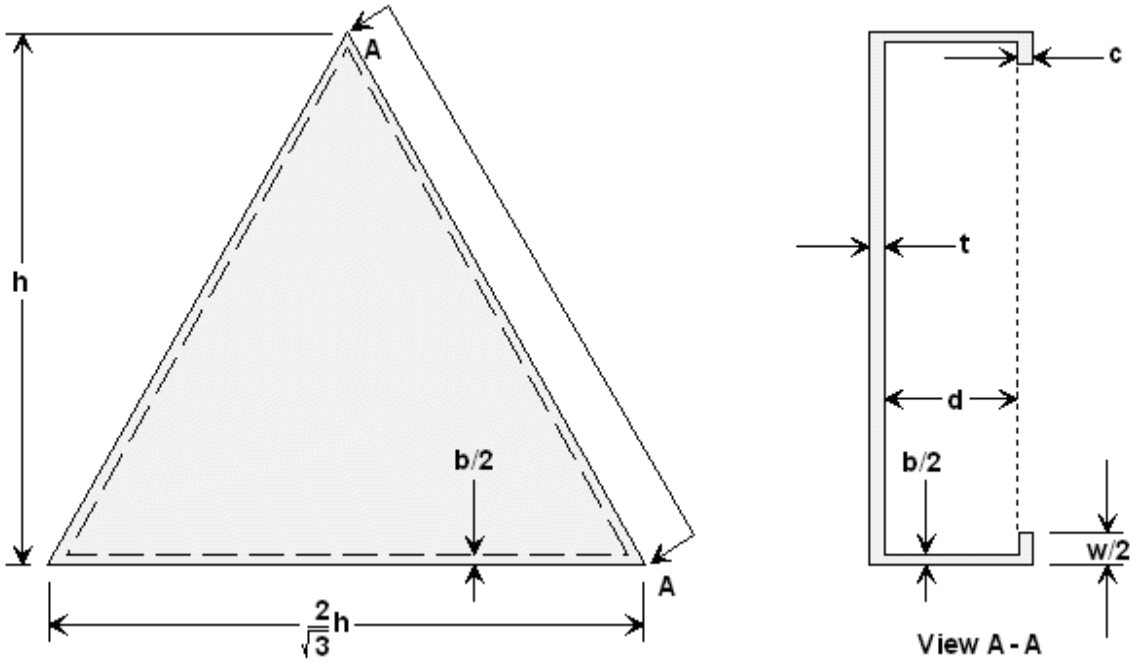
c - flange thickness

d - rib height

h - cell height

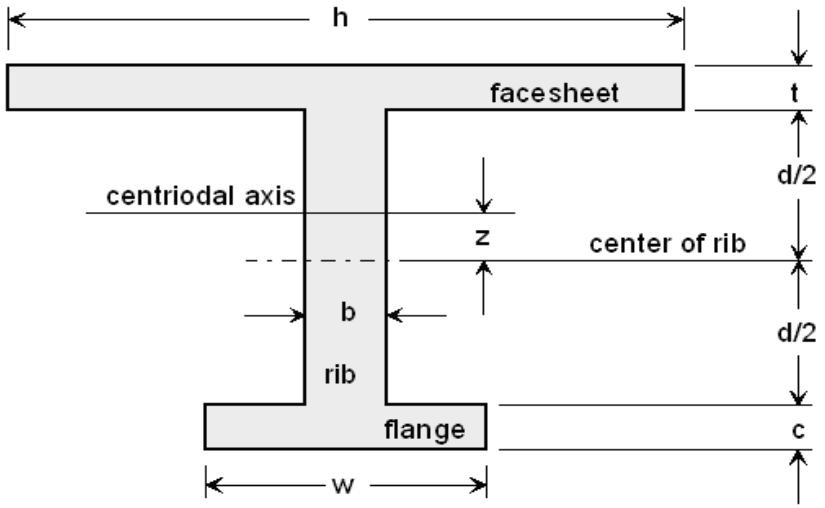
t - face sheet thickness

w - flange width



With these values it is possible to calculate the overall section properties of the isogrid plate and NASTRAN PSHELL input values. The derivation of the necessary equations is not included here. If you are interested in the math, please contact Dr. William Case at the e-mail address provided at the top and bottom of this page.

A Microsoft Excel spreadsheet is provided to calculate the section properties and PSHELL input values. (This spreadsheet will work with Excel versions 5, 6, and 7, and may work with other versions as well.) In the spreadsheet the values shown in the figure below are required (except z, which is called zbar in the spreadsheet).



If you have a plugin that reads Excel spreadsheets, then clicking on the link will launch the plugin; if you want to save the file to your hard drive, then hold down the shift key and click on the link.

Download the MS Excel spreadsheet ISOGRID.xls. (26k)

Dr. William Case, retired
Jeff Bolognese

Modal Weight Calculations

For translational DOF

$$\% = \frac{Apparent\ Weight}{Total\ Weight} \times 100$$

where

$$Apparent\ Weight = \left(PF_T\right)^2 \left(M_i\right) g$$

Participation factor :

$$\left(PF_T\right)_{xyz} = -\left(\frac{\Sigma F_{xyz}}{\lambda_i}\right) \frac{1}{M_i}$$

For rotational DOF

$$\% = \frac{Apparent\ Inertia}{Total\ Inertia} \times 100$$

where

$$Apparent\ Inertia = \left(PF_R\right)^2 \left(M_i\right) g$$

Participation factor:

$$\left(PF_R\right)_{xyz} = -\left(\frac{\Sigma m_{xyz}}{\lambda_i}\right)\frac{\mathbf{1}}{M_i}$$

$\boldsymbol{i}$ = mode number

$\boldsymbol{\lambda}$ = eigenvalue

$\boldsymbol{M}$ = generalized mass

$\boldsymbol{g}$ = 386.4 in/sec^2

$\boldsymbol{F_{xyz}}$ = constraint force in the x, y, or z direction

$\boldsymbol{m_{xyz}}$ = constraint moment about the x, y, or z axis

Total Inertia$_{xyz}$ = I_{xx}, I_{yy}, or I_{zz} about the mounting point

Method for Recovering Modal Mass Participation Output

MSC/NASTRAN version 2001 and beyond has the capability to extract modal mass participation directly from nomal modes analysis. No DMAP is required and it can be used with the normal modes solution sequence. The following is a sample executive and case control deck for a normal modes run with modal mass extraction. The cards that are specific for the modal mass extraction are shown in bold.

Sample Input

```
ID XYZ,MODALWT
SOL 103
TIME 10 APP DISP CEND
TITLE=XYZ TEST VERIFIED MODEL (english - renumbered) (08/96)
SUBTITLE=FIXED BASE DYNAMICS W/ASET
SPC=1
echo=none
SPCFORCE=ALL
METHOD=100
MEFFMASS(ALL)=YES
BEGIN BULK
```

Sample Output

M O D A L E F F E C T I V E M A S S S U M M A R Y

TOTAL EFFECTIVE MASS FRACTION
REFERENCE POINT AT ORIGIN OF BASIC COORDINATE SYSTEM

T1	T2	T3	R1	R2	R3
3.948519E-01	7.700374E-01	9.790530E-01	9.802241E-01	5.374004E-01	4.438883E-01

(This is the total percentage of mass represented in the number of modes extracted in the EIGR card)

EFFECTIVE MASS MATRIX

1.301470E-02	-1.445757E-05	6.120819E-04	-1.217158E-02	7.967635E-01	2.766647E-01
-1.445757E-05	2.536148E-02	-3.641499E-04	3.149053E-03	-8.146847E-04	-3.397166E-04
6.120819E-04	-3.641499E-04	3.227054E-02	-6.531930E-01	2.772698E-02	1.234497E-02
-1.217158E-02	3.149053E-03	-6.531930E-01	1.323802E+01	-5.505087E-01	-2.455033E-01
7.967635E-01	-8.146847E-04	2.772698E-02	-5.505087E-01	4.954358E+01	1.663506E+01
2.766647E-01	-3.397166E-04	1.234497E-02	-2.455033E-01	1.663506E+01	6.176721E+00

A-SET RIGID BODY MASS MATRIX

3.296097E-02	9.154070E-21	-5.196499E-20	9.980221E-19	1.717233E+00	6.654800E-01
9.154070E-21	3.293540E-02	-6.146909E-22	-5.571844E-05	3.835558E-19	1.002346E-04
-5.196499E-20	-6.146909E-22	3.296097E-02	-6.654800E-01	-9.742095E-05	-9.906851E-19
9.980221E-19	-5.571844E-05	-6.654800E-01	1.350510E+01	1.926013E-03	-2.541951E-05
1.717233E+00	3.835558E-19	-9.742095E-05	1.926013E-03	9.219119E+01	3.458480E+01

```
*   6.654800E-01  1.002346E-04 -9.906851E-19 -2.541951E-05  3.458480E+01  1.391504E+01 *
***                                                                                   ***
```

MODAL EFFECTIVE MASS FRACTION (***Translations Only***)

MODE NO.	FREQUENCY	T1 FRACTION	T1 SUM	T2 FRACTION	T2 SUM	T3 FRACTION	T3 SUM
1	3.443978E+01	1.087428E-07	1.087428E-07	2.878337E-01	2.878337E-01	4.671222E-04	4.671222E-04
2	9.164602E+01	1.037630E-07	2.125058E-07	3.013015E-03	2.908467E-01	1.752554E-01	1.757225E-01
3	9.314177E+01	1.176210E-06	1.388716E-06	8.141640E-06	2.908548E-01	8.531422E-05	1.758078E-01
4	9.319604E+01	6.164517E-09	1.394880E-06	4.916736E-07	2.908553E-01	6.009478E-05	1.758679E-01
5	1.190980E+02	5.566692E-04	5.580641E-04	7.114702E-05	2.909265E-01	3.210024E-07	1.758682E-01
6	1.246129E+02	4.152403E-05	5.995881E-04	1.206140E-01	4.115404E-01	2.999131E-06	1.758712E-01
7	1.332887E+02	1.709978E-07	5.997591E-04	1.782257E-01	5.897661E-01	3.270761E-04	1.761983E-01
8	1.387781E+02	1.097554E-07	5.998688E-04	9.015527E-02	6.799214E-01	1.928480E-04	1.763912E-01
9	1.720068E+02	5.168283E-06	6.050371E-04	3.729035E-07	6.799218E-01	8.354106E-07	1.763920E-01

(etc...)

MODAL EFFECTIVE MASS FRACTION (***Rotations Only***)

MODE NO.	FREQUENCY	R1 FRACTION	R1 SUM	R2 FRACTION	R2 SUM	R3 FRACTION	R3 SUM
1	3.443978E+01	5.643318E-04	5.643318E-04	1.352309E-07	1.352309E-07	1.171721E-07	1.171721E-07
2	9.164602E+01	1.997144E-01	2.002787E-01	3.633747E-07	4.986056E-07	2.032208E-07	3.203930E-07
3	9.314177E+01	9.733624E-05	2.003761E-01	1.086912E-06	1.585518E-06	1.360715E-06	1.681108E-06
4	9.319604E+01	6.838387E-05	2.004444E-01	2.265038E-09	1.587783E-06	1.896154E-09	1.683004E-06
5	1.190980E+02	2.877842E-07	2.004447E-01	5.968428E-04	5.984306E-04	1.837002E-02	1.837171E-02
6	1.246129E+02	8.937288E-06	2.004537E-01	3.818008E-05	6.366107E-04	6.438512E-05	1.843609E-02
7	1.332887E+02	4.437017E-04	2.008974E-01	7.682432E-07	6.373789E-04	7.551849E-07	1.843685E-02
8	1.387781E+02	2.582847E-04	2.011557E-01	5.440199E-09	6.373844E-04	3.157543E-08	1.843688E-02
9	1.720068E+02	6.662045E-07	2.011563E-01	5.183653E-06	6.425680E-04	5.199454E-06	1.844208E-02

(etc...)

MODAL PARTICIPATION FACTORS

MODE NO.	FREQUENCY	T1	T2	T3	R1	R2	R3
1	3.443978E+01	-5.986876E-05	-9.736486E-02	-3.923876E-03	8.730038E-02	-3.530878E-03	-1.276892E-03
2	9.164602E+01	5.848186E-05	-9.961669E-03	-7.600386E-02	1.642304E+00	5.787913E-03	1.681614E-03
3	9.314177E+01	-1.968985E-04	-5.178302E-04	-1.676914E-03	3.625652E-02	-1.001018E-02	-4.351368E-03
4	9.319604E+01	1.425442E-05	-1.272535E-04	-1.407403E-03	3.038965E-02	4.569645E-04	1.624347E-04
5	1.190980E+02	-4.283498E-03	1.530770E-03	-1.028618E-04	1.971435E-03	-2.345712E-01	-5.055883E-01
6	1.246129E+02	1.169903E-03	6.302753E-02	3.144110E-04	-1.098631E-02	5.932847E-02	2.993194E-02
7	1.332887E+02	-7.507498E-05	-7.661549E-02	-3.283405E-03	7.740952E-02	-8.415774E-03	-3.241670E-03
8	1.387781E+02	-6.014686E-05	5.449128E-02	2.521202E-03	-5.906065E-02	-7.081938E-04	-6.628523E-04
9	1.720068E+02	4.127368E-04	1.108229E-04	1.659396E-04	-2.999526E-03	2.186063E-02	8.505915E-03

(etc...)

MODAL EFFECTIVE MASS

MODE NO.	FREQUENCY	T1	T2	T3	R1	R2	R3
1	3.443978E+01	3.584268E-09	9.479916E-03	1.539680E-05	7.621356E-03	1.246710E-05	1.630454E-06
2	9.164602E+01	3.420128E-09	9.923486E-05	5.776588E-03	2.697162E+00	3.349994E-05	2.827825E-06
3	9.314177E+01	3.876902E-08	2.681481E-07	2.812040E-06	1.314536E-03	1.002037E-04	1.893440E-05

```
4     9.319604E+01        2.031885E-10        1.619346E-08        1.980782E-06        9.235310E-04        2.088166E-07
2.638505E-08
5     1.190980E+02        1.834836E-05        2.343255E-06        1.058055E-08        3.886554E-06        5.502365E-02
2.556195E-01
6     1.246129E+02        1.368672E-06        3.972469E-03        9.885427E-08        1.206990E-04        3.519867E-03
8.959212E-04
7     1.332887E+02        5.636253E-09        5.869933E-03        1.078075E-05        5.992234E-03        7.082525E-05
1.050842E-05
8     1.387781E+02        3.617645E-09        2.969299E-03        6.356459E-06        3.488161E-03        5.015384E-07
4.393732E-07
9     1.720068E+02        1.703516E-07        1.228172E-08        2.753594E-08        8.997157E-06        4.778871E-04
7.235059E-05
      (etc...)
```

The type of information presented by the MEFFMASS card can be modifed. Check the MSC user manual for information on the parameters available for this card.

Preloads

Applying Preloads with the Differential Stiffness Approach

Differential Stiffness Theory

In geometric non-linear problems, displacements are large. Therefore, the equilibrium equations must be written with respect to the deformed geometry. This theory assumes that the geometric stiffness matrix can be added to the structural (or differential) stiffness matrix; therefore, the change in geometry of the structure is only reflected in the stiffness matrix.

Differential Stiffness Matrix

A matrix that is necessary to account for the change in potential energy associated with rotation of continuum elements under load. (stiffness effects that depend linearly on displacements)

Fundamental Assumption

External loads do not change in magnitude or direction as the structure deflects. However, the loads do travel with the grid points.

Properties of KDGG (Differential Stiffness Matrix)

1. Symmetric
2. Independent of elastic properties
3. Depends on element geometry, displacement field, and state of stress
4. Indefinite and cannot be inverted

Setting Up the NASTRAN Data Deck

Using the differential stiffness approach to applying preloads to a NASTRAN finite element model consists of two executions of the model. The first model is set up to obtain the differential stiffness matrix (KDGG) as a results of the preload. The second model is where the differential stiffness matrix is added to the geometric stiffness matrix and desired results are requested.

The DMAPs shown are for use in MSC/NASTRAN Version 67.

The First Model

Executive Control

```
Sol 5
Compile sol5, souin=mscsou
Alter 313
Output4 KDGG//-1//11 $(creates a fort.11 file of KDGG)
Exit $
```

Case Control

```
Title = 'title'
Subtitle =
SPC = 'boundary points'
$'Output requests' - probably want to output elements in which the
$ preload is being applied to verify the magnitude.
Subcase 1
Deform = 'id'
Subcase 2
```

Bulk Data

Add the necessary number of 'DEFORM' bulk data cards. The 'DEFORM' card requests the deformation of a one dimensional element, i.e. BAR, ROD, TUBE. For a preload on a bar, the deformation is:

delta = PL / EA

where P is the applied preload. You can verify that the correct preload has been applied to the element by reviewing the BAR, ROD, or TUBE, element forces.

The Second Model

Executive Control

```
Sol 24
Compile sol24, souin=mscsou
Alter 90
Inputt4 /KDGG,,,,/1/11/-1 $ Add KGG, KDGG/K1X $
Equiv K1X,KGG/Always $
Endalter
```

Case Control and Bulk Data decks should be set up as any other run would be.

NOTE: The differential stiffness matrix can be very big. Therefore, you should be careful to ensure that the matrix is not sent across the network. This will slow the job execution substantially. If this is a problem, use the ASSIGN card to write the matrix to the scratch directories on the computer that is executing the NASTRAN job.

Cherie Congedo
May 18, 1995

Random Vibration

Nastran Input Deck Example

Following is an example of an MSC.Nastran v. 2001 random vibration analysis run. All the NASTRAN cards necessary to perform a random run are described here. Comments follow the card (or cards) being described. The random-specific cards are in bold. The model is a simple cantilever beam with 48 elements and 49 nodes. It lies along the Y-axis with the first node, 101 at the origin. The downloadable data file can be found at the bottom of this page.

NOTE: This method will only work for MSC's version 2001 or later of NASTRAN because they (finally!) did away with the large mass needed to do frequency response runs. This method is extremely similar to UAI's version of frequency response runs. (See notes below for the differences.)

Input Data Deck

```
$EXECUTIVE CONTROL DECK
ID BEAM,RANDOM
SOL 111
```

Modal Frequency Response Solution Number

```
$
TIME 10 CEND
$
$CASE CONTROL DECK
TITLE = RANDOM VIBRATION EXAMPLE MODEL
SUBTITLE = 48-ELEMENT CANTILEVER BEAM ON THE X-AXIS
LABEL = Y-DIRECTION RANDOM INPUT OF 0.01 G^2/HZ FROM 20-2000HZ
$ Beam is along the X-axis and is 48 inches long
$
SPC     = 1
METHOD  = 1
DLOAD   = 10
SDAMPING = 20
RANDOM  = 30
FREQ    = 40
```

SPC refers only to the SPC1 card (not the SPCD card);
METHOD refers to EIGR card;
DLOAD refers to RLOADx card;
SDAMPING refers to TABDMP1 card;
RANDOM refers to RANDPS card;
FREQ refers to all FREQx cards used

```
$
ECHO = NONE
$
$SET 99 = 101
$SPCFORCES(PHASE) = 99
```

The preceding two cards are used for shaker point force recovery only (remove $ if needed). SET 99 sets the shaker GRID 101 output;
SPCFORCES requests force recovery at SET 99 (see OUTPUT and SPCD cards below)

```
$
OUTPUT(XYPLOT)
XYPEAK,XYPUNCH,ACCE,PSDF/ 101(T2),149(T2)
```

Cards used for plot output.
NOTE: THIS IS THE ONLY WAY TO GET RANDOM OUTPUT!
OUTPUT card is required to use XYPLOT commands;

XYPEAK recovers RMS and other peak responses in the .prt file (required for RMS output);
XYPUNCH creates a .pch file of response vs. frequency;
ACCE requests acceleration output (FORCE requests force output);
PSDF requests output as PSD;
GRID point 101 and 149 responses recovered; T1=X, T2=Y, T3=Z, as well as R1, R2, and R3

```
$
BEGIN BULK
$ PARAM,AUTOSPC,YES
PARAM,GRDPNT,0
PARAM,RESVEC,YES
```

PARAM,RESVEC,YES is REQUIRED for accurate results when NOT using the seismic mass. It computes residual vectors.

```
GRID       101     0       0.     0.      0.      0
```

Shaker grid point. Often this is an RBE element attached to many GRID points with the independent GRID being the shaker point.

```
$
SPC1    1 123456101
```

The Shaker point is SPC'd in all DOF

```
$
RLOAD2,10,11,,,12,,ACCE
```

RLOAD defines frequency response dynamic loading;
LOAD = SPCD * TABLED1 [= A * B(f)];
11 IS SPCD card;
12 IS TABLED1 card;

ACCE refers to type of dynamic excitation, enforced acceleration in this case.
Other options are LOAD for applied load, DISP for enforced displacement, and VELO for enforced velocity.

```
$
SPCD,11,101,2,1.0
```

Y-axis input at GRID 101;
2 refers to input direction (1=X, 2=Y, 3=Z);
1.0 is any scale factor;

```
$
$SPCD,11,101,2,386.4
```

Use this card instead when recovering forces or displacements (assuming you're using the inch/pound system), such as at the shaker point. If you're using a metric based system, such as mm/gram, then 386.4 becomes 9807. (You can also change the 1.0 values in the TABLED1 card below to the acceleration value and get the same result. But do not change both.)

```
TABLED1      12    LOG      LOG                                          +
+          20.0    1.0  2000.0    1.0     ENDT
```

TABLED1 is a tabular function defining the frequency-dependent portion of RLOAD2
The table is given as X1,Y1, X2,Y2, X3,Y3, etc.;
20.0 is the beginning frequency (X1); 1.0 is the magnitude (Y1);
2000.0 is the ending frequency (X2); 1.0 is the magnitude (Y2);
The Y values are used as scale factors; the actual Y values are entered as random input in TABRND1 below (not frequency input);
The LOG LOG entries refer to how the data is interpolated on each axis.

NOTE: All tables MUST end with an ENDT card.
Also, tables are not always compatible between UAI/NASTRAN and MSC/NASTRAN. The tables used in the older version of this example would not work under Nastran 2001.

```
TABDMP1   20                                                          +
+       20.0    0.1     2000.0     0.1  ENDT
```

Defines modal damping of Q=10;
The damping is 1/Q over the range of frequencies; in this case Q is constant. Q can vary over frequency if the proper data is available.

```
$
RANDPS,30,1,1,1.0,0.0,31
```

Defines random PSDF loading;
The important value here is the 31, which refers to the TABRND1 card

```
TABRND1  31     LOG     LOG                                           +
+       19.999  0.01    20.0    0.01   2000.0   0.01   2000.1   0.01  +
+       ENDT
```

Defines PSDF from 20 to 2000 Hz on a log-log scale; First value must be slightly lower than actual frequency; Format is Frequency, PSD,
Final value must be slightly higher than actual frequency

```
$
FREQ1,40,20.0,2.0,140
FREQ1,40,300.0,5.0,140
FREQ1,40,1000.0,20.0,50
FREQ3,40,20.0,2000.0
```

FREQx cards input frequencies where responses are recorded;
FREQ1 gives beginning frequency, freq. increment size and number of increments;
FREQ3 adds each natural frequency FROM 20 TO 2000 Hz

```
$
EIGRL         1     0.    2000.              0                   MASS
```

EIGRL determines normal modes from 0 to 2000 Hz

```
$
INCLUDE 'randbulk.dat'
```

Bulk Data can be elsewhere in a different file.

```
$ ENDDATA
```

Eigenvalue Output

The eigenvalue and frequency table from the .F06 file is given below. It gives the frequency range requested in the EIGRL card, 0-2000Hz. Note that the frequencies listed in this table under CYCLES correspond with the peaks in the plot at the bottom of the page.

SINE VIBRATION EXAMPLE MODEL
48-ELEMENT CANTILEVER BEAM ON THE X-AXIS
Y-DIRECTION SINE INPUT

R E A L E I G E N V A L U E S

MODE NO.	EXTRACTION ORDER	EIGENVALUE	RADIANS	CYCLES	GENERALIZED MASS	GENERALIZED STIFFNESS
1	1	5.673224E+03	7.532081E+01	1.198768E+01	1.000000E+00	5.673224E+03
2	2	5.673224E+03	7.532081E+01	1.198768E+01	1.000000E+00	5.673224E+03
3	3	2.220890E+05	4.712632E+02	7.500386E+01	1.000000E+00	2.220890E+05
4	4	2.220890E+05	4.712632E+02	7.500386E+01	1.000000E+00	2.220890E+05
5	5	1.733352E+06	1.316568E+03	2.095384E+02	1.000000E+00	1.733352E+06
6	6	1.733352E+06	1.316568E+03	2.095384E+02	1.000000E+00	1.733352E+06
7	7	6.614887E+06	2.571942E+03	4.093373E+02	1.000000E+00	6.614887E+06
8	8	6.614888E+06	2.571942E+03	4.093373E+02	1.000000E+00	6.614888E+06
9	9	1.793594E+07	4.235084E+03	6.740346E+02	1.000000E+00	1.793594E+07
10	10	1.793594E+07	4.235085E+03	6.740347E+02	1.000000E+00	1.793594E+07
11	11	3.965145E+07	6.296939E+03	1.002189E+03	1.000000E+00	3.965145E+07
12	12	3.965146E+07	6.296940E+03	1.002189E+03	1.000000E+00	3.965146E+07
13	13	4.175332E+07	6.461681E+03	1.028408E+03	1.000000E+00	4.175332E+07
14	14	7.650954E+07	8.746973E+03	1.392124E+03	1.000000E+00	7.650954E+07
15	15	7.650957E+07	8.746975E+03	1.392124E+03	1.000000E+00	7.650957E+07
16	16	1.339317E+08	1.157289E+04	1.841882E+03	1.000000E+00	1.339317E+08
17	17	1.339318E+08	1.157289E+04	1.841882E+03	1.000000E+00	1.339318E+08

Example random output of peak values.

The PSD output for GRIDs 101(T2) and 149(T2) from the run's .F06 file is given below. The output is considered to be in units of G2/Hz, while the Grms value is in G. The PSD values are actually (in/sec2)2/Hz, but because frequency is dimensionless, we can pretend that G and in/sec2 are equal, i.e., an input of 1 G comes out the same as an input of 1 in/sec2.

If force, displacement or stress output is desired however, the acceleration must be multiplied by 386.4 in/sec2 in the SPCD card to obtain pounds. (If you are using metric units of mm, then you must multiply by 9807 mm/sec2.)

The Grms values are highlighted in the fourth column. (MSC could probably do a better job of formatting their output tables!) Also of note are the minimum and maximum Y values (G2/Hz) and their corresponding X values (frequency). You will probably have to scroll to the right to see everything.

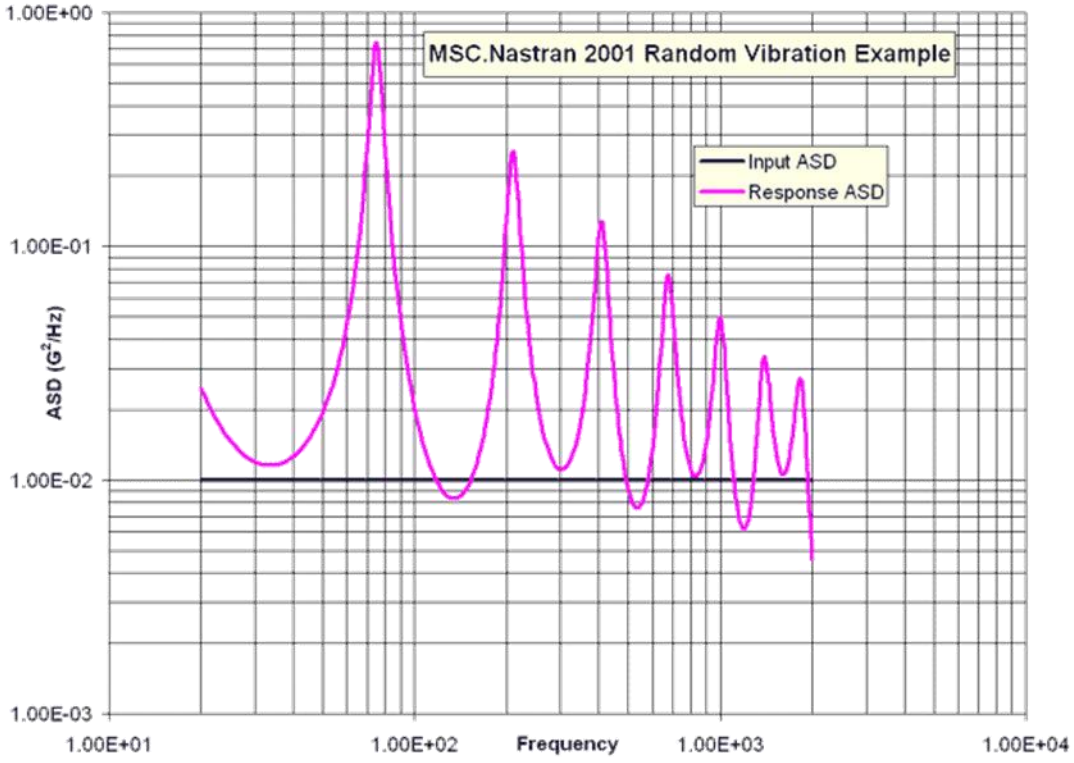
RANDOM VIBRATION EXAMPLE MODEL
48-ELEMENT CANTILEVER BEAM ON THE X-AXIS
Y-DIRECTION RANDOM INPUT OF 0.01 G^2/HZ FROM 20-2000HZ

X Y - O U T P U T S U M M A R Y (A U T O O R P S D F)

PLOT TYPE	CURVE TYPE	CURVE ID./ PANEL : GRID ID	RMS VALUE	NO. POSITIVE CROSSINGS	XMIN FOR ALL DATA	XMAX FOR ALL DATA	YMIN FOR ALL DATA	X FOR YMIN	YMAX FOR ALL DATA	X FOR YMAX
PSDF	ACCE	101(4)	**4.449719E+00**	1.160517E+03	2.000E+01	2.000E+03	**1.000E-02**	**5.667E+01**	**1.000E-02**	**2.000E+01**
PSDF	ACCE	149(4)	**7.309907E+00**	9.355332E+02	2.000E+01	2.000E+03	**4.553E-03**	**2.000E+03**	**7.485E-01**	**7.500E+01**

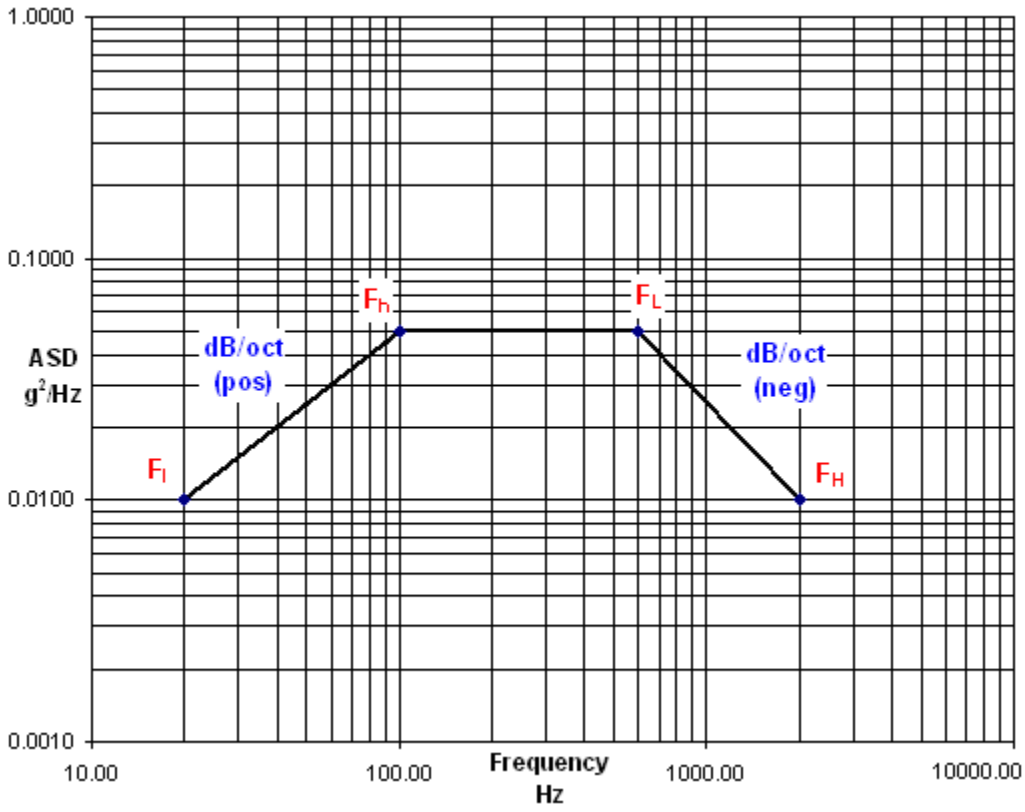
Plot of the Example Random Vibration Data

Below is a plot of the data given in the punch file (random_test.pch). It shows the input and the response. The input is a straight line at .01 G2/Hz. The response shows the peaks corresponding to the modes of the beam.



Specification Magnitude Equations

When performing a random vibration analysis, an input spec is generally given in a form such as the log-log plot in the figure or written in the table below. The problem is what to do with such information. We cannot input these values directly into NASTRAN because it will not accept a slope in dB/Oct. Individual points in G2/Hz vs. Hz are required. This page details how to get from the input to graphical points needed.



(Note that regardless of popular opinion, G2/Hz is actually an acceleration spectral density (ASD), not a power spectral density (PSD). PSD refers to the actual plot generated during testing, which simply reads the power output from the accelerometers.)

In tabular form, the input may be given in this form (beginning and ending frequencies are not always necessary if a continuous line is assumed):

Random Input Spec	
3.01 dB/Oct	
100.00 Hz	0.0500 G^2/Hz
600.00 Hz	0.0500 G^2/Hz
-4.02 dB/Oct	

Because no beginning or ending frequencies, Fl and FH, are given in this table, they must first be decided upon. This is generally project specific. However, the frequency range is usually 20Hz to 2000Hz. From the graph, Fl = 20Hz and FH = 2000Hz. Fh and FL are 100Hz and 600Hz, respectively.

First, determine the number of octaves between the two frequencies. Keep in mind that an octave is the doubling of the frequency. So going from 1Hz to 2Hz is an octave and going from 1000Hz to 2000Hz is also an octave. Thus, the number of octaves could be estimated from the graph above. The equation to calculate the exact number is:

$$\#Octaves = \frac{\log\left(F_H / F_L\right)}{\log(2)}$$

where FH is the higher frequency and FL is the lower frequency.

Second, determine the number of dB by multiplying the number of octaves by the slope, making sure to use the correct sign (positive or negative) for the slope:

$$dB = \left(\frac{dB}{Oct}\right) \times \left(\#Oct\right) = 10\log\left(\frac{ASD_H}{ASD_L}\right)$$

The previous equation also shows the definition of dB, where ASDH and ASDL are the acceleration spectral densities for the higher and lower frequencies respectively (NOT for the higher and lower ASD values! That is, ASDL can be greater than ASDH whereas FH is always greater than FL).

Finally, solve for the ASD at the desired frequency:

$$\frac{ASD_H}{ASD_L} = 10^{\left(dB/10\right)}$$

Or, for those of you who want a more expanded and complete version (where m is the slope in dB/Oct):

$$\frac{ASD_H}{ASD_L} = 10^{\left[\left(\frac{m}{10}\right)\left(\frac{\log\left(F_H/F_L\right)}{\log(2)}\right)\right]} = \left(\frac{F_H}{F_L}\right)^{\frac{m}{10\log(2)}}$$

And a final, simplified version:

$$\frac{ASD_H}{ASD_L} \approx \left(\frac{F_H}{F_L}\right)^{\frac{m}{3.01}}$$

I'll leave it up to you to use the example given in the graph above to prove that the equations work.

Thanks to Bob Coladonato and Dr. Bill Case, both retired from Goddard, and Jaap Wijker of University of Technology Delft in the Netherlands for their assistance with this page.

Calculating GRMS

It is very easy to describe the Grms (root-mean-square acceleration, sometimes written as GRMS or Grms or grms or grms) value as just the square root of the area under the ASD vs. frequency curve, which it is. But to physically interpret this value we need to look at Grms a different way. The easiest way to think of the Grms is to first look at the mean square acceleration.

Mean-square acceleration is the average of the square of the acceleration over time. That is, if you were to look at a time history of an accelerometer trace and were to square this time history and then determine the average value for this squared acceleration over the length of the time history, that would be the mean square acceleration. Using the mean square value keeps everything positive.

The Grms is the root-mean-square acceleration (or rms acceleration), which is just the square root of the mean square acceleration determined above.

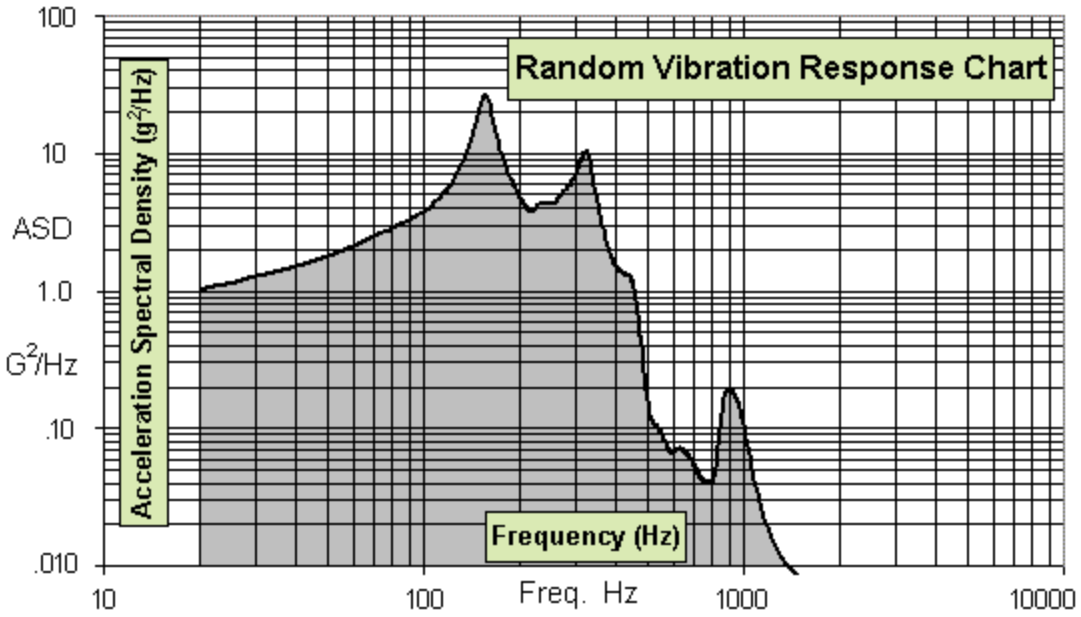
If the accelerometer time history is a pure sinusoid with zero mean value, e.g., a steady-state vibration, the rms acceleration would be .707 times the peak value of the sinusoidal acceleration (if just a plain average were used, then the average would be zero). If the accelerometer time history is a stationary Gaussian random time history, the rms acceleration (also called the 1 sigma acceleration) would be related to the statistical properties of the acceleration time history (you may have to refresh your probability and statistics knowledge for this):

- 68.3% of the time, the acceleration time history would have peaks that would not exceed the +/- 1 sigma accelerations.
- 95.4% of the time, the acceleration time history would have peaks that would not exceed the +/- 2 sigma accelerations.
- 99.7% of the time, the acceleration time history would have peaks that would not exceed the +/- 3 sigma accelerations.

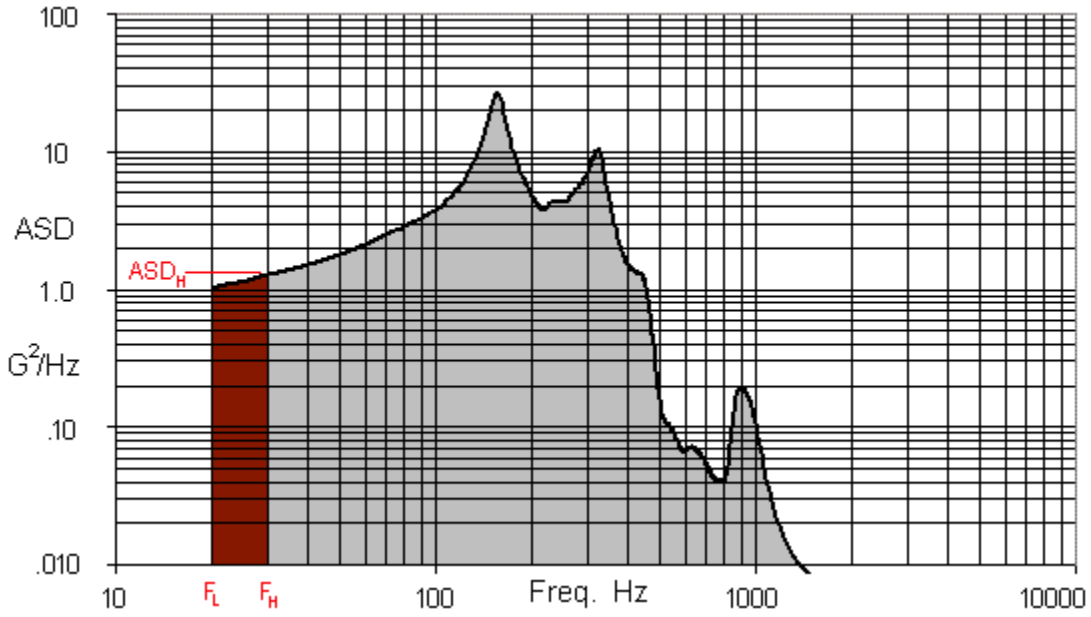
There is no theoretical maximum value for the Gaussian random variable; however, we typically design to 3 sigma since it would only be theoretically exceeded 0.3% of the time. In addition, from a practical point of view, we know that it would be physically impossible to achieve unreasonably high sigma values.

Below is presented the method to calculating the root-mean-square acceleration (Grms) response from a random vibration ASD curve.

Typical random vibration response curve:



Grms values are determined by the square root of the area under a ASD vs. frequency response curve. The Acceleration Spectral Density values are in g2/Hz and the frequencies are in Hz.



The figure above shows a bandwidth of 10 Hz, which will be used as an example for calculating Grms.

First, calculate the number of octaves. From the plot, FL = 20Hz and FH = 30Hz. The equations below gives #octaves = 0.58 .

$$\#Octaves = \frac{\log\left(F_H / F_L\right)}{\log(2)}$$

Second, calculate the dB value. For FL = 20Hz, ASDL = 1.0 g^2/Hz, while ASDH = 1.1 g^2/Hz. The calculated value is 0.41 dB. (The definition of dB is also provided.)

$$dB = 10\log\left(ASD_H / ASD_L\right)$$

Third, calculate the slope, m, of the segment between the frequencies FL and FH. Dividing the number of dB by the number of octaves gives m = 0.71 dB/oct.

$$m = dB / \#octaves$$

Fourth, calculate the area under the curve between the frequencies FL and FH. In our example, A = 10.53 g^2.

$$A = 10\log(2)\frac{ASD_H}{10\log(2)+m}\left[F_H - \left(F_L\right)\left(\frac{F_L}{F_H}\right)^{m/10\log(2)}\right]$$

NOTE - The above equation is invalid if the slope m = -10log(2) because you would be dividing by zero. If m = -10log(2), use the following equation for the area:

$$A = ASD_L \times F_L \times \ln\left(\frac{F_H}{F_L}\right)$$

(L'hôpital's Rule is used to solve the equation as the limit of [1 + m/(10log2)] goes to zero -- or some such nonsense.)

Finally, take the square root of the area for the Grms value. To finish our example, the acceleration = 3.24 Grms.

$$g_{rms} = \sqrt{A}$$

In order to calculate the Grms value for the entire curve, sum up all the areas (A1 + A2 + A3 + ... + An = A) and take the square root of the sum.

NOTE: 3dB is a factor of 2 for ASD curves (g^2/Hz) while 6dB is a factor of 2 for Grms values. For example, reducing a peak ASD value of 12g^2/Hz by -3dB would give you 6g^2/Hz; reducing a value of 12 Grms -3dB results in a value of 9 Grms and reducing it -6dB results in a value of 6 Grms. This tends to be confusing for people new to random vibration.

Thanks to Bob Coladonato and Bill Case, both now retired from Goddard, and Jaap Wijker of University of Technology Delft in the Netherlands for their assistance with this page.

Creating a Test Specification

One purpose for performing a random vibration analysis is to create a component test level specification. With a finite element model, a random vibration analysis can be performed to predict acceleration responses from 20 Hz to 2000 Hz. This response is in turn used as a template to derive a test level specification. Use these guidelines to create the spec.

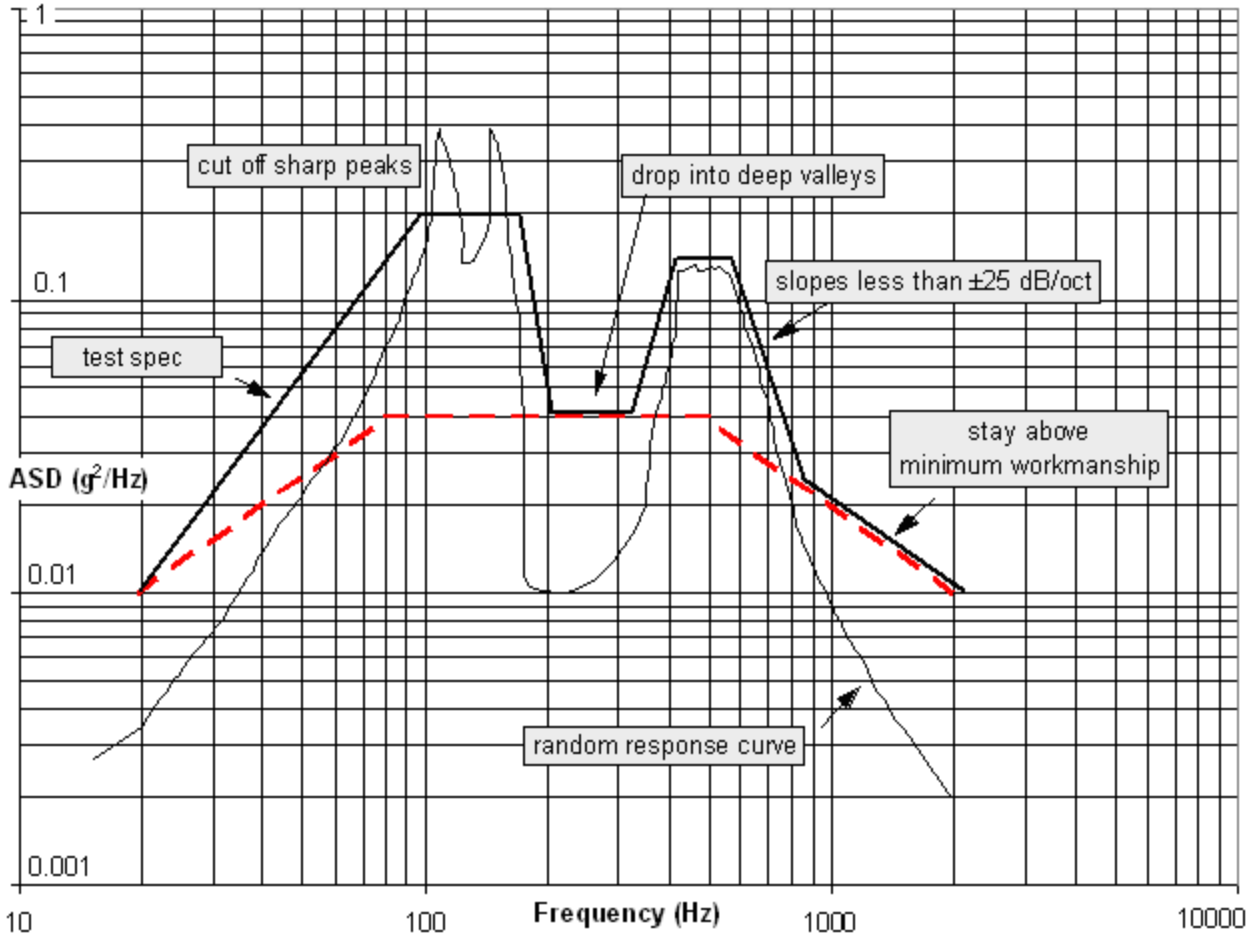
1. From NASTRAN random vibration response data (or whatever analysis software you're using), create a log-log plot of the response of ASD (acceleration spectral density) vs. frequency. **NOTE** This will work using any type of ASD response curve. For example, if an engineering model (EM) is being tested, responses from accelerometers on the EM can also be used.

2. Add in Component Minimum Workmanship to the plot. For components less than 100 lbs (45.4 kg), NASA uses these values from GEVS, Table 2.4-5:

Minimum Workmanship	
Frequency	**ASD**
20 Hz	0.01 g^2/Hz
20-80 Hz	+3 dB/oct.
80-500 Hz	0.04 g^2/Hz
500-2000 Hz	-3 dB/oct.
2000 Hz	0.01 g^2/Hz
Overall	**6.8 g_{rms}**

3. Enclose response curve inside test spec curve using the following rules:
   - Slopes should be less than + 25 dB/oct or greater than - 25 dB/oct (this depends on your shaker table, consult your test engineer for exact slope limits)
   - Frequency bands should be greater than 10 Hz.
   - Don’t plot spec below Minimum Workmanship
   - Sharp peaks can be cut off at about 1/2 their height (-3 dB) *e.g., a sharp peak of 0.52 g2/Hz can be cut off at > 0.26 g2/Hz
   - Drop spec curve into valleys that are large
   - Try to keep the overall grms level of the spec to be no more than 1.25 times the overall grms level of the response curve.

The following plot is an example of a spec to be used for component random vibration testing.



* NOTE: 3dB (3.01dB to be exact) is a factor of 2 for ASD curves (g^2/Hz) while 6dB is a factor of 2 for grms values. For example, reducing a peak ASD value of 12g^2/Hz -3dB would give you 6g^2/Hz; reducing a value of 12grms -3dB results in a value of 8.48grms and reducing it -6dB results in a value of 6.0grms. This tends to be confusing for people new to random vibration.

Thanks to Bob Coladonato, retired from Goddard, for his assistance with this page.

Miles Equation

Ryan Simmons
NASA Goddard Space Flight Center
May 2001

The following equation is attributed to John W. Miles.

$$G_{RMS} = \sqrt{\frac{\pi}{2} f_n Q \, [ASD_{input}]}$$

Where

G_{RMS} = Root Mean Square Acceleration in G's (sometimes given as $\ddot{y}_{RMS}$).

f_n = Natural Frequency

$Q = \frac{1}{2\zeta}$ = Transmissibility (or amplification factor) at f_n

where ζ is the critical damping ratio.

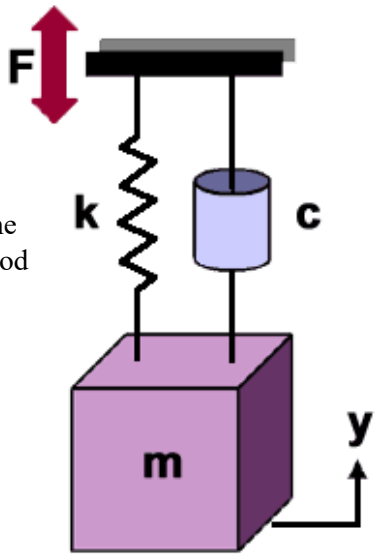
$[ASD_{input}]$ = Input Acceleration Spectral Density at f_n in units of $\frac{g^2}{Hz}$.

In 1954, Miles developed his version of this equation for GRMS as he was researching fatigue failure of aircraft structural components caused by jet engine vibration and gust loading. Miles simplified his research by modeling a system using one degree of freedom only. He also applied statistical advances that had been made at the time. While his goal was to analyze the stress of a component, the equation can be rearranged and used to determine, among others, displacement, force, and, in our case, acceleration.

Single Degree of Freedom System

Miles' Equation is derived using a single degree of freedom (SDOF) system (lightly damped), consisting of a mass, spring and damper, that is excited by a constant-level "white noise" random vibration input from 0 Hz to infinity. Miles' Equation is thus technically applicable only to a SDOF system.

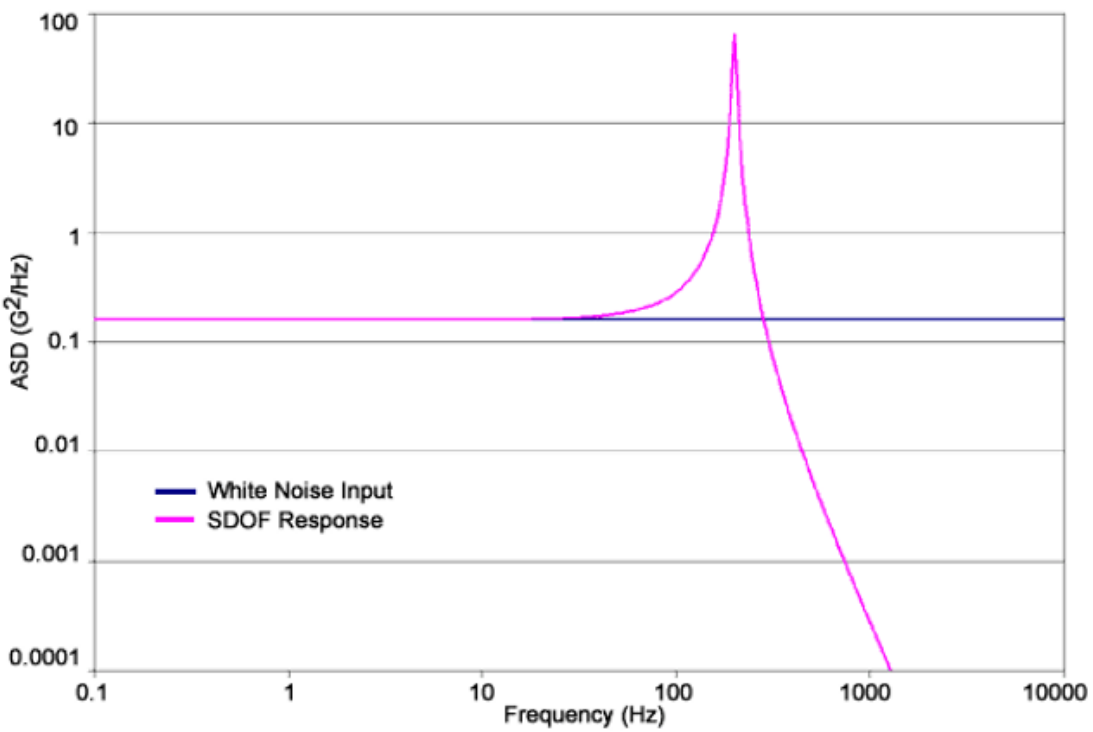
This figure shows a typical representation of a SDOF oscillator. The mass (m) is attached to the spring (stiffness k) and the damper (damping c). The system is forced by the random vibration function (F) in the y-direction only.



Derivation of Miles' Equation is left up to you. The references at the bottom of the page provide different methods of deriving the equation, so reviewing each method could provide insight into Miles' Equation.

SDOF System Response Plot

The plot below shows the input and response of a SDOF system. Miles' Equation calculates the square root of the area under the response curve, providing us with the GRMS value.



SDOF System Response Plot

Response Parameters

Miles' Equation can also be used to predict other response parameters such as stress or displacement. For example, the displacement equation for YRMS is given here. Displacement can in turn be used to calculate the stress.

$$Y_{RMS} = \sqrt{\frac{Q[ASD_{input}]}{32\pi^3(f_n)^3}}$$

NOTE: In this form of the equation, since we want units of length, the ASDinput must be in units of $[length/sec^2]^2/[Hz]$.

You will most likely have to convert your ASD data from g2/Hz to something else.

USES OF MILES' EQUATION

Design

During the design of a part, if enough analysis has been performed to determine the part has a predominant resonant frequency, then Miles' Equation can be used to estimate the loads due to random vibration. Just calculate the GRMS value and multiply it by 3. That's the "three-sigma" load. Keep in mind, this is conservative.

Testing

Accelerations due to random vibration at resonant frequencies in a multiple degree of freedom system can be approximated using Miles' Equation. This will indicate how much of the overall RMS acceleration is occurring at a resonant peak of interest compared to the complete frequency spectrum.

MISUSES OF MILES' EQUATION

Miles' Equation Does NOT Work in Reverse

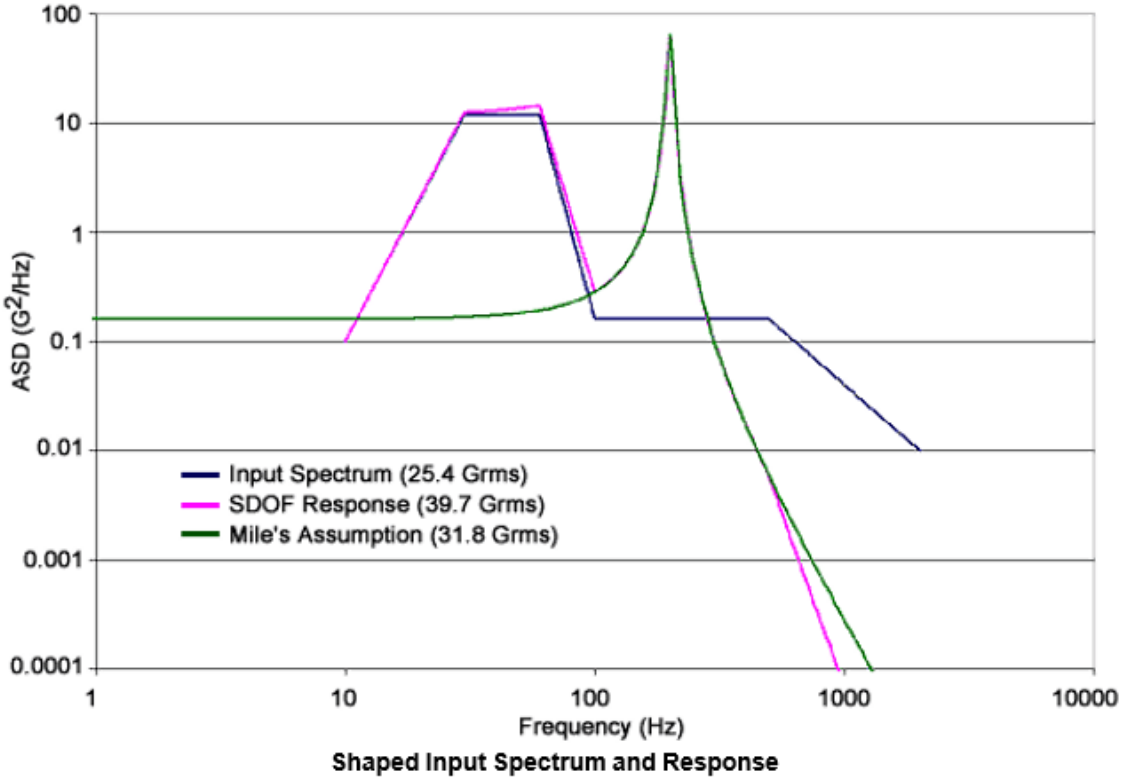
Accelerations cannot be determined during random vibration testing using Miles' Equation. An upper bound on loads can be calculated using the 3-sigma value, but that's about it. In other words, a part designed to 3-sigma equivalent static loads will survive a random vibration test. However, Miles' Equation cannot be used to predict the failure of a part designed to less than 3-sigma levels.

Miles' Equation Does NOT Give an Equivalent Static Load

Calculating the GRMS value at a resonant peak after a random vibration test and multiplying it by the test article mass does not mean that the test article was subjected to that same, equivalent static load. It simply provides a statistical calculation of the peak load for a SDOF system. The actual loading on a multiple DOF system due to random input depends on the response of multiple modes, the mode shapes and the amount of effective mass participating in each mode. Static testing must still be done.

Miles' Equation may not be Conservative for a Shaped Input Spectrum

Miles' Equation is based on the response of a SDOF system subjected to a flat random input. If Miles' Equation is used to calculate the GRMS response to a shaped input spectrum, the result will not accurately reflect the rigid body response below the resonant frequency. If the shaped input spectrum has high ASD levels below the resonant frequency, then Mile's Equation may significantly underpredict the GRMS response. The following plot shows this condition. Note that the Miles' assumption has a response of 31.8 GRMS while the actual SDOF response is much greater at 39.7 GRMS. This demonstrates that Miles' Equation is best used when the input ASD is flat or nearly so.



Shaped Input Spectrum and Response

REFERENCES

John W. Miles, On Structural Fatigue Under Random Loading, Journal of the Aeronautical Sciences, pg. 753, November, 1954.

Dave S. Steinberg, Vibration Analysis for Electronic Equipment, Wiley-Interscience, New York

Thomas P. Sarafin (editor), Spacecraft Structures and Mechanisms: From Concept to Launch, Microcosm, Inc., Torrance, CA, 1995

Thanks to Scott Gordon and Bob Coladonato for their invaluable assistance in the creation of this page.

Ryan Simmons
May 2001

Definition of Decibels

The decibel is a unitless method of expressing the ratio of two quantities. The expression is in terms of the logarithm to base 10 of the ratio instead of the raw ratio. This is done for convenience in expressing the ratio of numbers many magnitudes apart with decibel numbers that are not as large.

If G1 and G2 are two vibration acceleration amplitudes, then their decibel ratio (or decibel difference) is:

$$dB_{12} = 20\,LOG_{10}\frac{G1}{G2} \qquad 1.$$

For example, if G1 is twice G2, then

$$dB_{12} = 20\,LOG_{10}\,2 = \mathbf{6.0206}\ dB$$

Or G1 is about 6 dB greater than G2. If G1 were half as large as G2 then:

$$dB_{12} = 20\,LOG_{10}\,0.5 = -\mathbf{6.0206}\,dB$$

Or G1 is about 6 dB less than G2. It is seen from the dB equation that each factor of 10 in the ratio results in 20 dB difference in amplitudes.

Note from equation 1 that:

$$\boxed{dB_{12} = 10\,LOG_{10}\frac{G1^2}{G2^2}} \qquad 2.$$

That is, for squares of amplitudes (power) each factor of 10 in the ratio results in 10 dB difference in power. Thus, a factor of 1000 in the power ratio is a 30 dB difference in power. Equation 2 is the normal way of defining decibels, and equation 1 results from it.

Also for convenience, decibels are used rather than Bels. From Equation 2, a Bel would be defined as:

$$B_{12} = LOG_{10} \frac{G1^2}{G2^2}$$

Seeing the equation written out in this form makes the definition much clearer.

Thanks to Dr. Bill Case for the definition.

Sine Vibration

Nastran Input Deck Example

Following is an example of a NASTRAN swept sine frequency response analysis run. All the NASTRAN cards necessary to perform a frequency response run are described here. Comments follow the card (or cards) being described. The frequency response-specific cards are in bold. The model is a simple cantilever beam with 48 elements and 49 nodes. It lies along the Y-axis with the first node, 101 at the origin. The downloadable data file can be found at the bottom of this page.

NOTE: This method will only work for MSC's version 2001 or later of NASTRAN because they (finally!) did away with the large mass needed to do frequency response runs. This method is extremely similar to UAI's version of frequency response runs. (See notes below for the differences.)

```
$EXECUTIVE CONTROL DECK
ID SINE,TEST
SOL 111
```

Modal Frequency Response Solution Number

```
$
TIME 10
CEND
$
$CASE CONTROL DECK
TITLE = SINE VIBRATION EXAMPLE MODEL
SUBTITLE = 48-ELEMENT CANTILEVER BEAM ON THE X-AXIS
LABEL = Y-DIRECTION SINE INPUT
$ Beam is along the X-axis and is 48 inches long

$
ECHO = NONE
$
SPC      = 1
METHOD   = 1
DLOAD    = 10
SDAMPING = 20
FREQ     = 40
```

METHOD refers to EIGRL card;
DLOAD refers to RLOADx card;
SDAMPING refers to TABDMP1 card;
FREQ refers to all FREQx cards used

```
$
$SET 98 = 101
$SPCFORCES(PHASE) = 98
```

For force recovery, these two cards create SET1 for the shaker point, GRID 101, then request forces (as magnitude/phase) at the shaker point (see SPCD card below)

```
$
SET 99 = 101, 149
```

Defines GRIDs for response output

```
$
ACCELERATION(SORT2,PRINT,PUNCH,PHASE) = 99
```

Requests frequency response output for SET 99;
Acceleration output given as magnitude/phase angle;
Output is written to the .F06 and punch files;
SORT2 sorts output as GRID1, all frequencies; GRID2, all frequencies; etc.
(SORT1 sorts output as frequency 1, all GRIDs; frequency 2, all GRIDs; etc.)

```
$
OUTPUT(XYPLOT)
XYPEAK,XYPUNCH,ACCE/ 101(T2),149(T2)
```

Cards used for plot output.
XYPEAK recovers peak responses in the .prt file;
XYPUNCH creates a .pch file of response vs. frequency;
ACCE requests acceleration output (FORCE requests force output);
GRID point 101 and 149 responses recovered; T1=X, T2=Y, T3=Z, as well as R1, R2, and R3
(NOTE: the punch file will contain both ACCELERATION and OUTPUT data.)

```
$
BEGIN BULK
$
PARAM,AUTOSPC,YES
PARAM,GRDPNT,0
PARAM,RESVEC,YES
```

PARAM,RESVEC,YES is REQUIRED for accurate results when NOT using the seismic mass. It computes residual vectors.

```
$
GRID         101       0      0.      0.      0.       0
```

Shaker grid point.

```
$
SPC1           1  123456     101
```

The Shaker point is constrained in all DOF. Often this is an RBE element attached to many GRID points with the independent GRID being the shaker point.

```
$
RLOAD2,10,11,,,12,,ACCE
```

RLOAD defines frequency response dynamic loading;
LOAD = SPCD * TABLED1 [= A * B(f)];
11 IS SPCD card;
12 IS TABLED1 card;
ACCE refers to type of dynamic excitation, enforced acceleration in this case.

```
$
SPCD,11,101,2,1.0
```

Y-axis input at GRID 101;
2 refers to input direction (1=X, 2=Y, 3=Z);

1.0 is any scale factor;

```
$
$SPCD,11,101,2,386.4
```

Use this card instead when recovering forces or displacements, such as at the shaker point. If you're using the Metric system, for example mm, then use 9807 instead of 386.4.

```
$
$
TABLED1       12     LOG     LOG                                        +
+           2.0    0.25   500.0    0.25    ENDT
```

TABLED1 is a tabular function defining the frequency-dependent portion of RLOAD2
This table gives the sine vibration input spec;
The table is given as X1,Y1, X2,Y2, X3,Y3, etc.;
2.0 is the beginning frequency (X1); 0.25 is the magnitude (Y1);
500.0 is the ending frequency (X2); 0.25 is the magnitude (Y2);
If your input is written as double amplitude (D.A.), a conversion is available.

All tables MUST end with an ENDT card.

```
$
TABDMP1   20                                                          +
+       20.0    0.1     2000.0     0.1   ENDT
```

Defines modal damping of Q=10;
The damping is 1/Q over the range of frequencies; in this case Q is constant.

```
$
FREQ1,40,2.0,2.0,250
FREQ3,40,2.0,500.0
```

FREQx cards input frequencies where responses are recorded;
FREQ1 gives beginning frequency, freq. increment size and number of increments;
FREQ3 adds each natural frequency up to 500 Hz

```
$
EIGRL         1     0.     500.              0                   MASS
```

EIGRL determines normal modes up to 500 Hz

```
$
INCLUDE 'sinebulk.dat'
```

Bulk Data can be elsewhere in a different file.

```
$
ENDDATA
```

Double Amplitude

When you input your frequency and acceleration data into a sine vibration frequency response NASTRAN deck, you may run into that cryptic notation, 0.32 inch D.A. from 10-20Hz. This is what is known as a vibrational amplitude. But since NASTRAN does not know what D.A. is, and you probably don't either, you need to convert that value into something everyone understands, such as a vibrational acceleration, which could be measured in g's or in/sec2 or m/s2. Fortunately, this conversion is not difficult to do.

Start with the fact that acceleration is the second time derivative of length:

$$x = x_o \sin \omega t$$

$$\dot{x} = x_o \omega \cos \omega t$$

$$\ddot{x} = -x_o \omega^2 \sin \omega t$$

At maximum amplitude (ignore the minus sign):

$$\sin \omega t = 1$$

Therefore:

$$\ddot{x}_{max} = x_o \omega^2$$

By definition x_o is the single amplitude:

$$x_o = \frac{D.A.}{2}$$

Also, we know:

$$\omega = 2\pi f$$

Finally we get the acceleration in terms of the length you are using (length/sec2):

$$\ddot{x} = \left(\frac{D.A.}{2} \right) (2\pi)^2 (f)^2$$

$$\ddot{x} = 19.74 f^2 D.A.$$

If you are working in an inch/pound system and you want your answer in g, here is the final equation:

$$\ddot{x} = 0.0511 f^2 D.A.$$

Semi-Rigid Connections

One of the basic assumptions of conventional structural analysis is that joints are either perfectly rigid or perfectly hinged. Therefore, when analyzing a structure, joints are idealized as either fixed or hinged. However, in actual structures, typical connections do not behave in either a perfectly rigid or a perfectly hinged manner. The different types of connections that are commonly used fill the entire flexibility spectrum from flexible "hinge like" connections to semi-rigid connections to rigid connections. These non-ideal connections are often referred to as flexible connections, semi-rigid connections, spring hinged joints, or sticky hinges.

To achieve a more accurate analysis of a structure it would be advantageous to include the true behavior of the joints. For example, there is a substantial variation in the distribution of bending moment for a beam with hinged, semi-rigid, and fixed joints (Figure 1). (W = force/unit length)

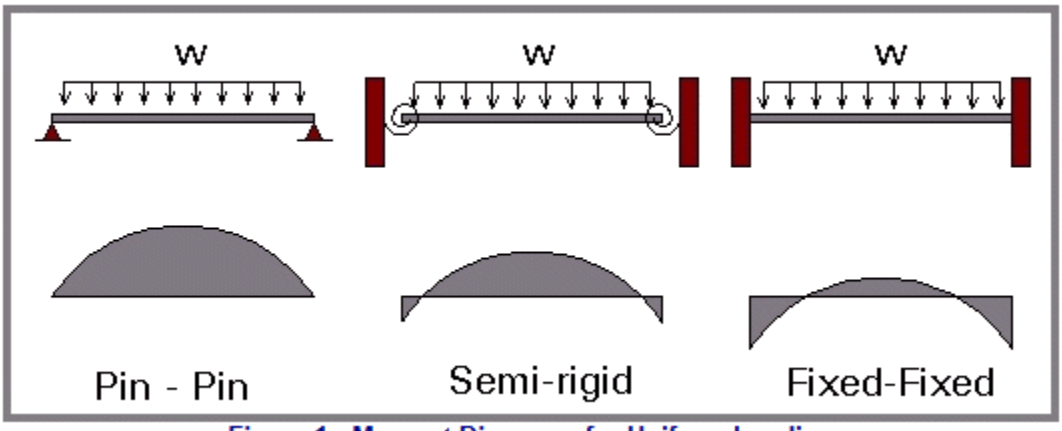


Figure 1 - Moment Diagrams for Uniform Loading

There is also a substantial difference in the minimum natural frequency and mode shape of a beam with hinged, semi-rigid, and fixed joints (Figure 2).

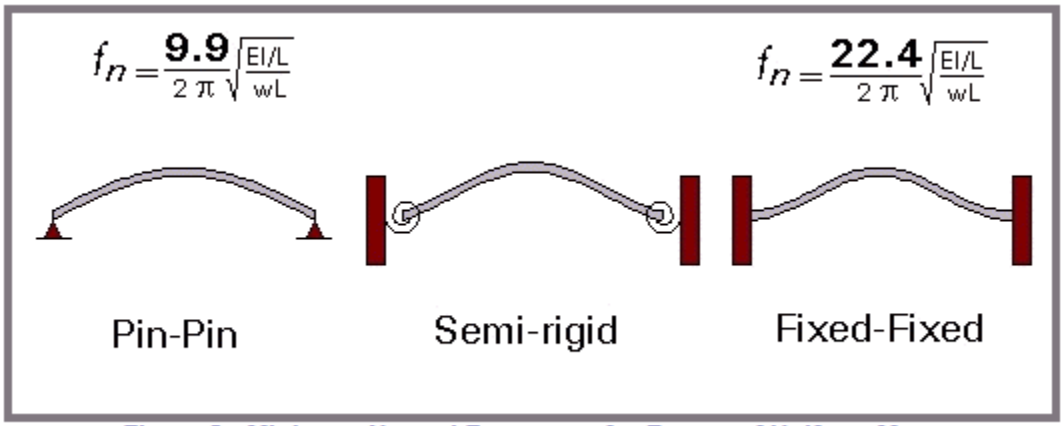


Figure 2 - Minimum Natural Frequency for Beams of Uniform Mass

E = Young's Modulus; I = moment area of inertia; L = length of beam; W = force/unit length

Semi-rigid joints can be modeled using a rotational spring which separates a member from it's surrounding environment. The rotational spring is characterized by it's moment-rotation curve, where the rotational stiffness of the spring k is equal to the slope of the moment-rotation curve (Figure 3}. The value of k for a given connection detail can be determined by testing.

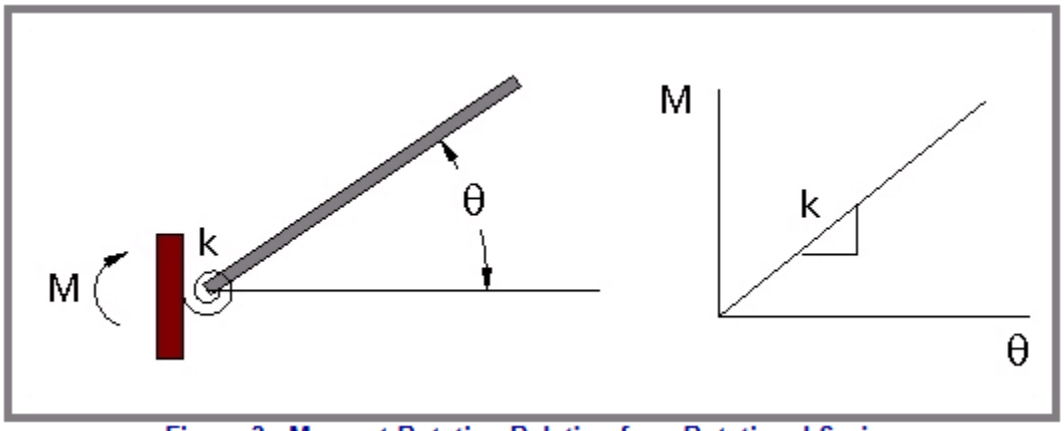


Figure 3 - Moment-Rotation Relation for a Rotational Spring

The magnitude of the stiffness of the spring hinge k does not determine if a joint will behave in a rigid or flexible manner. Rather it is the ratio of the spring stiffness k to the flexural stiffness of the member EI/L that determines what behavior will be exhibited. This ratio is often called the "joint stiffness" and is defined as kL/EI.

In order to visualize this phenomenon consider some "medium sized" spring with spring stiffness K-medium (Figure 4). If a very flimsy, spaghetti-like member is connected to it, even though the spring is only "average sized" it will still keep the flimsy member from rotating at the support, just like a rigid connection would. However, if a very stiff member were connected to our "average sized" spring, it would appear as if the spring had little to no effect in restraining the member.

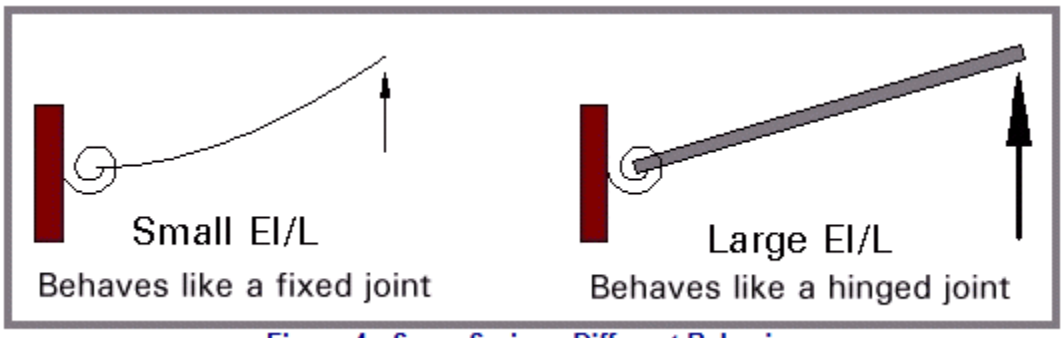


Figure 4 - Same Spring - Different Behavior

This now raises the question of what specific values of the stiffness ratio will result in hinged behavior, semi-rigid behavior, or fixed behavior. To answer this question the dynamic and static behavior of a prismatic beam with uniform load was investigated.

A finite element model of a typical sized beam with spring hinges at the supports was created (Figure 5 inset). A NASTRAN normal modes analysis was performed to obtain the minimum natural frequency for various values of the spring stiffness k. The results of this analysis are shown in Figure 5.

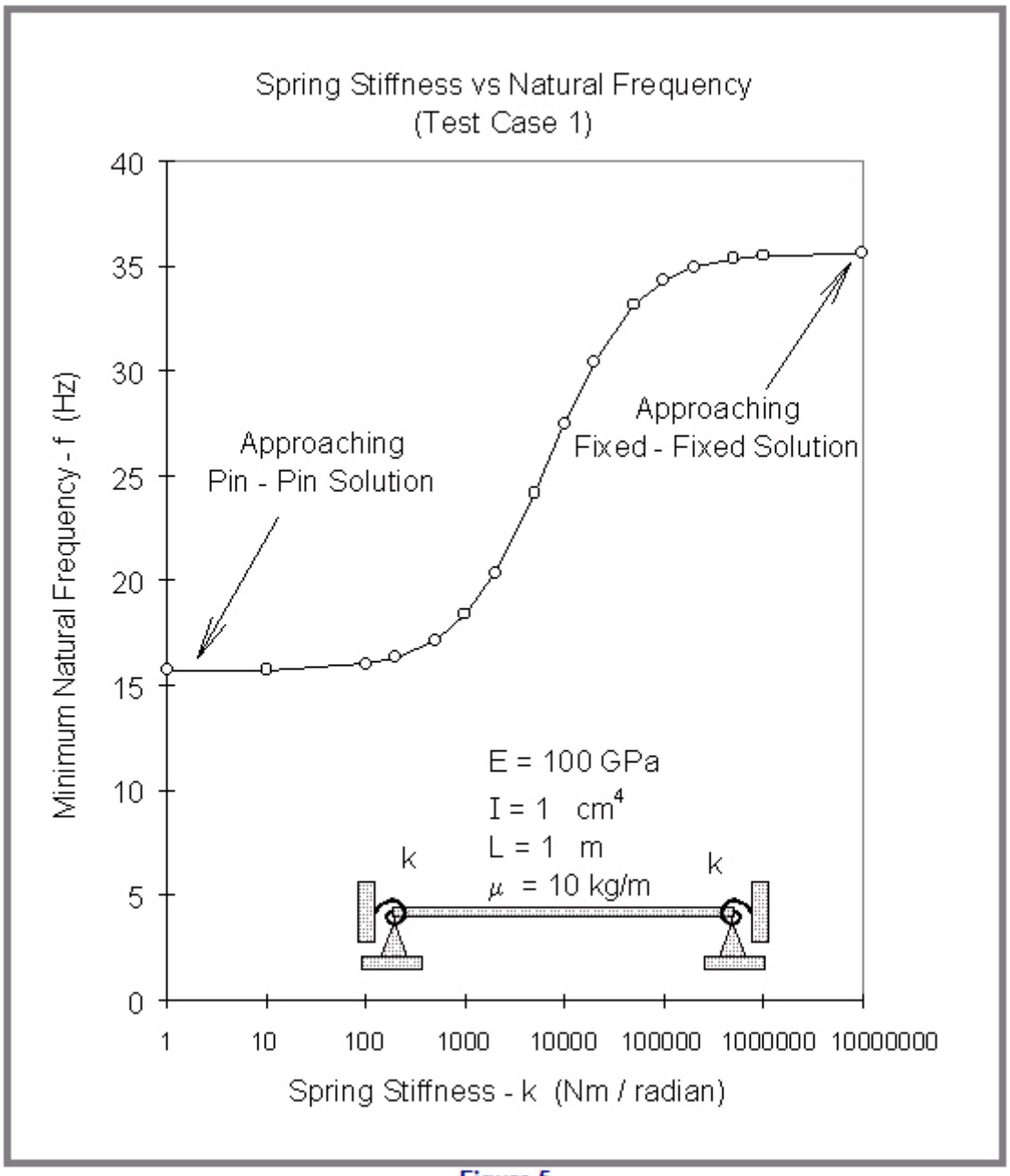


Figure 5

As k gets smaller the solution approaches the pin-pin solution. as k gets larger the solution approaches the fixed-fixed solution. Furthermore, there is a discernable range in which the behavior changes from pined to fixed.

The results shown in Figure 5 are for a specific beam. For the results to be of use they must be applicable to any beam. Therefore, the behavior of a second beam with increased mass (100 times greater than the initial test case) and the behavior of a third beam with increased flexural stiffness (100 times greater than the initial test case) was investigated, the results are shown in Figure 6.

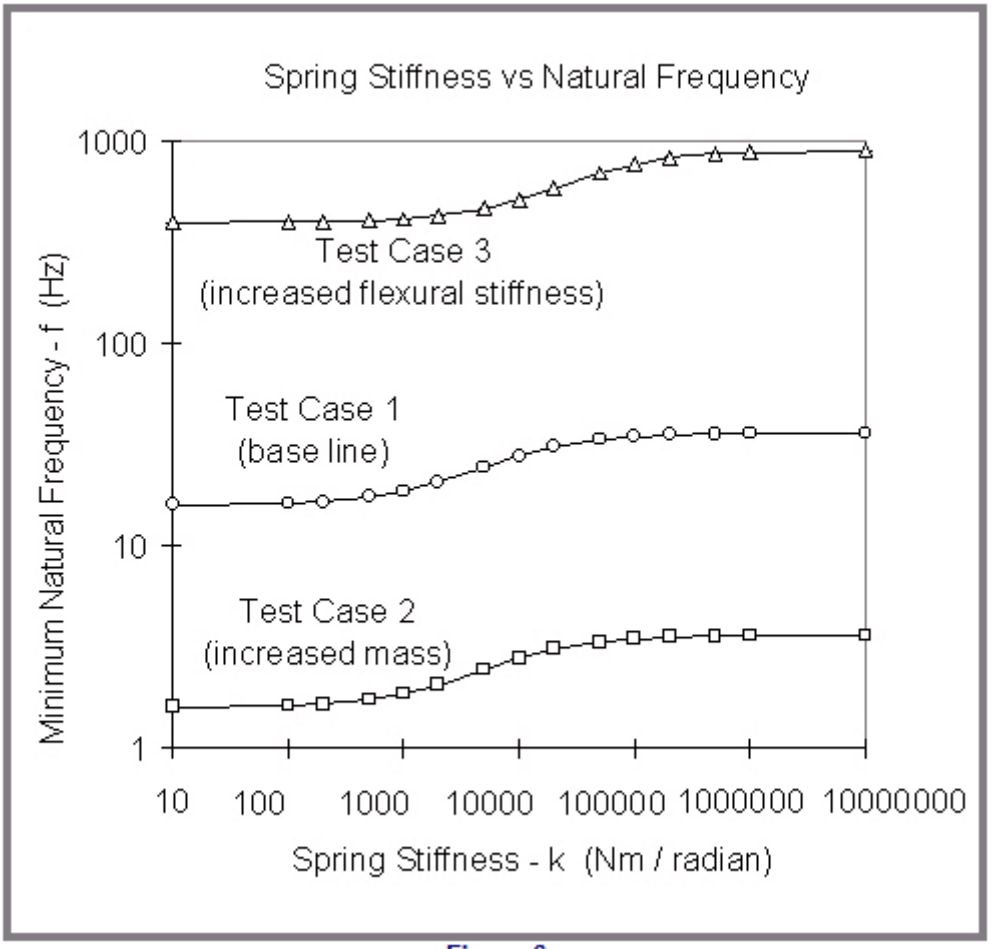


Figure 6

When the mass was increased the natural frequency dropped but the transition range remained the same. Therefore it appears that the mass has no effect on the joint stiffness. When the flexural stiffness was increased the natural frequency increased and the transition range shifted also, demonstrating that the flexural stiffness of the beam effects the joint stiffness.

The previous results can be non-dimensionalized. The natural frequency can be non-dimensionalized using the natural frequency of the pin-pin case as zero and the natural frequency of the fixed-fixed case as 1. And instead of the spring stiffness the joint stiffness ratio of K/ EI/L will be used. The results plotted in this manner are shown in Figure 7.

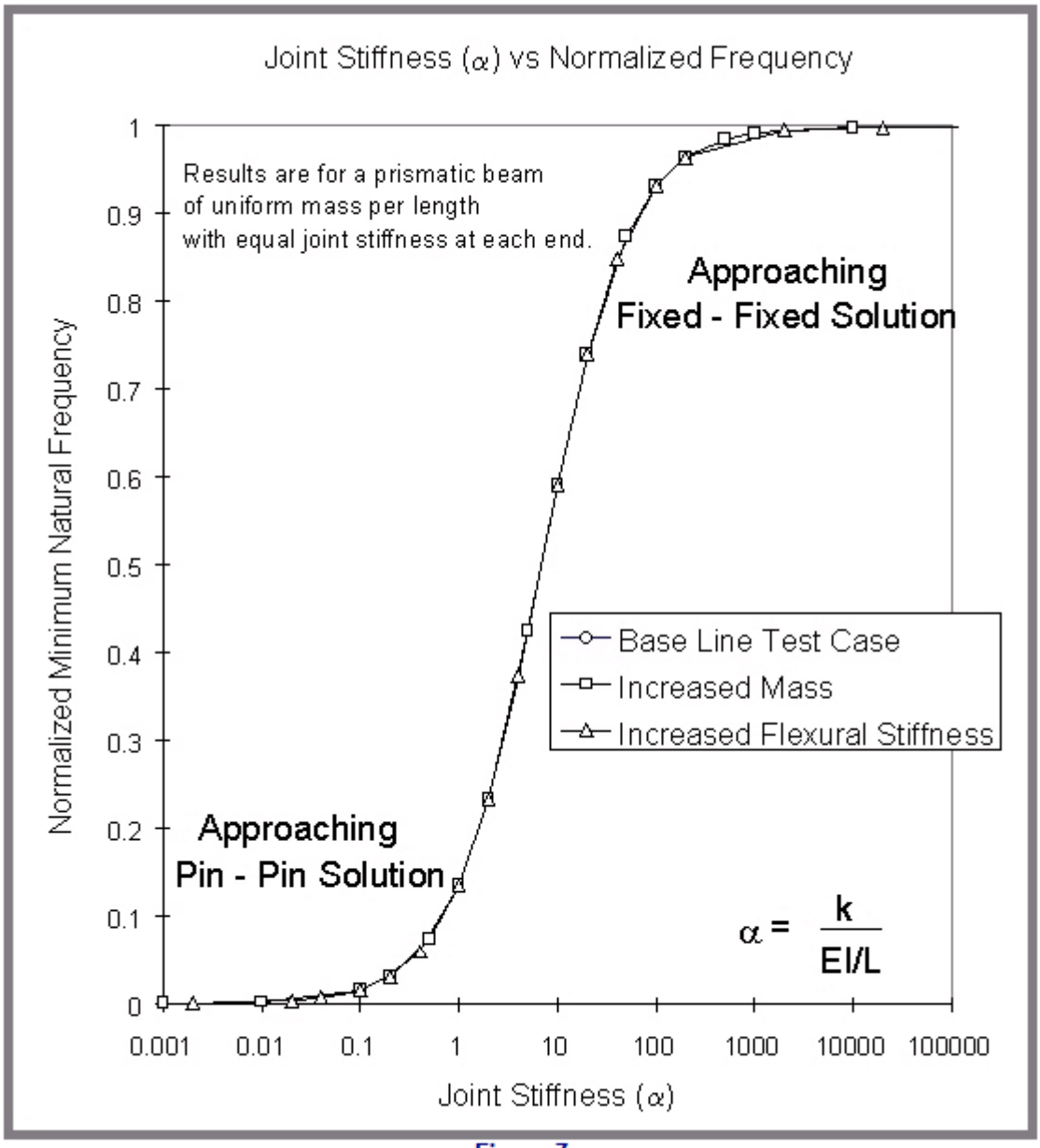


Figure 7

When plotted in this manner all three test cases fall on the same line. From this it can be seen that if the joint stiffness is less than 1 the natural frequency is within 10% of the pin-pin condition and if the joint stiffness is greater than 100 the natural frequency is within 10% of the fixed-fixed condition.

Conclusion

Rotational springs can be used to model semi-rigid restraint conditions. The restraint conditions can be characterized by a quantity called the joint stiffness. The joint stiffness depends on the ratio of the spring stiffness k to the flexural stiffness EI/L of the beam and is defined as kL/EI. NASTRAN analysis results confirm that the mass of a beam does not effect the joint stiffness, only the flexural stiffness of the beam does.

It was also found that when the joint stiffness is less than 1.0 the beam behaves as if it is hinged, while if the joint stiffness is greater than 100 the beam behaves as if it is fixed. Therefore it may not be worth the effort to model a connection with a rotational spring if the resulting joint stiffness being used is less than 1 or greater than 100. Furthermore, if semi-rigid behavior is desired a joint stiffness between 1 and 100 should be used. Figure 7 can help in determining what joint stiffness should be to used to achieve a particular behavior.

Written by Dr. John McGuire

Unit Consistency

Have you ever wondered why your results look a bit odd sometimes: are those pound forces or pound masses?!... will I get Newtons if I input millimeters?! You complete your analysis only to determine your units were wrong. If so, then this table of consistent units might help you.

Examples of Consistant Systems of Units for Structural Analysis											
System of Units	*Input*								*Output*		
	Length	Force	Elastic Modulus	Mass	Mass Density	WTMASS Param	Acceleration (1 G)		Disp	Force	Stress
1 Metric meter-kg	m	N	Pa	kg	Kg/m^3	1.0	9.807	m/sec^2	m	N	Pa
2 Metric mm-ton	mm	N	MPa	t or Mg	t/mm^3 or Mg/mm^3	1.0	9807	mm/sec^2	mm	N	MPa
3 English ft-lb	ft	lb_f	psf	slug	$slug/ft^3$	1.0	32.17	ft/sec^2	ft	lb_f	psf
4 English in-lb (mass)	in	lb_f	psi	lb_f - sec^2/in	lb_f - sec^2/in^4	1.0	386.1	in/sec^2	in	lb_f	psi
5 English in-lb (force)	in	lb_f	psi	lb_f	lb_f/in^3	0.00259	386.1	in/sec^2	in	lb_f	psi

NOTE:

t = Metric ton = 1000 kg = 1 Mega gram (Mg)

Validity Checks

After a finite element model is created and before results are used from that model, Code 542 performs several standard validity checks on the model. This document will describe these standard validity checks.

There are four mathematical validity checks. They are:

Unit Enforced Displacement and Rotation

Free-Free Dynamics with a Stiffness Equilibrium Check

Unit Gravity Loading

Unit Temperature Increase

The first three checks should be performed on all models used for either statics or dynamics analyses. For models developed for thermal distortion analyses, the Unit Temperature Increase check should also be performed.

Computer-Aided Accuracy Checks

In addition to the mathematical validity checks given above, there are many other model checks that can and should be performed. The checks listed below require a pre/post processor, such as FEMAP, Pro/Engineer, or Patran. Furthermore, you should be aware of any warnings or potential errors your solver gives you, such as grid point singularities. This is not an exhaustive list, just some suggestions.

Free Edge - Use a free-edge check to make sure elements are connected properly, particularly where 1D, 2D, and 3D elements are connected to zero-length elements (springs).

Coincident Nodes - Check for erroneous coincident nodes; but be careful not to merge intentional coincident nodes.

Coincident Elements - Check for coincident elements; this can happen if you are not careful when creating multiple meshes.

Bar/Beam Orientation - Use orientation visualization to determine if bar and beam elements are oriented properly.

Plate Element Distortion - Check elements for problems like aspect ratio, internal angles, warping.

Plate Element Normals - Check that plate normals are in the same direction where necessary.

Mass - Check to be certain the model's mass is reasonable and accurate.

Common Sense Accuracy Checks

The following checks should speak for themselves. They are just simple, common sense checks we should make on every model. If you've been bitten by a common sense check that's not on this list, e-mail it to us and we'll consider putting it on the list.

Material Properties - Make sure the property values are correct, match the units you're using elsewhere, and include all the information you need. For example, thermal expansion coefficient, reference temperature, density, etc. And know if you're using mass density or weight density. It could cause problems when retrieving forces.

Element Properties - Verify beam cross sections and corresponding area moment of inertias, plate thicknesses and element offsets. Verify mass and inertia properties for concentrated mass elements and stiffnesses for elastic elements.

Units - Are your units consistent? NASTRAN doesn't care what units you use and won't give you any warnings. Some pre-processors will also let you use inconsistent units. Take particular care with strange, mixed units (e.g. - mm and Kg). Are you using mass or weight units? You may need a PARAM,WTMASS card. For inch/pound models, the WTMASS value is 0.00259.

Dimensions - Check to make sure that your overall geometry is correct. Check your FEM vs. drawings or CAD. This is especially important if you've transformed units.

Grounding Conditions - Make sure your constraints are in the correct locations and DOFs. Check NASTRAN SPC forces to make sure only your mount points are listed.

Proper Element Use - Are you using the right element for the job? Are you using it correctly? For example, RBE3's aren't really "rigid" and shouldn't be used as structural elements. They are primarily for calculating average motion or distributing loads. CELAS elements should be between coincident grids or may generate artificial forces during rigid body rotation.

MPC's and RBE's - Are your MPC's formulated correctly? Are you using the appropriate RBE? For example, is that component you're modeling with that RBE2 really stiffer than the structure you're attaching it to? Will you artificially lock up your structure with a big RBE? Avoid over-constraining your structure with RBE's.

Coordinate Systems - If you're using multiple input and output coordinate systems, make sure that you know which ones your elements and constraints are referencing. For example, RBE elements will use the grid output coordinate system when assigning DOFs.

Understand singularities - If you have singularity errors, are they because of rotational degrees of freedom from plates and solids? Are those removed by AUTOSPC? If you are unsure, then you should try to remove the singularities and re-run the model. If some still exist, they may indicate grounding problems.

Validity checks, i.e., validation of the finite element model, are defined as checks that ensure the model is mathematically accurate. Validity checks do not ensure the accuracy of the model in representing a physical system, just that the model will give mathematically correct results.

Accuracy checks help the modeler to determine and ensure that the model represents the physical system he or she is modeling.

- Contrast these checks with the verification of a model, which is generally done by correlating a model with actual mechanical test results.

Unit Enforced Displacement and Rotation

Purpose

The Unit Enforced Displacement and Rotation check verifies that no illegal grounding (such as incorrect SPC or MPC) is present in the model. The model should move as a rigid body when it is translated one unit or rotated one radian. This check verifies displacements, element forces, and grid point forces. The check is performed in all three translational and all three rotational directions. This check outputs lots of data.

Output Results

The displacements results from the three translational subcases should be 1.0 in the input direction and 0.0 in the other five directions. From the rotational subcases, the rotation in the input direction should be 1.0 and the other two rotations should be 0.0. The three translational displacements from the rotational subcases will not be zero. The element forces and grid point force balances should be zero with a maximum of 0.1 lb force and 1.0 in-lb moment. The grid point held should not have other SPC's, and, for best results, should be near the center of gravity of the model.

Input Required

EXECUTIVE DECK must include the following information:

```
SOL 101
```

CASE CONTROL must include the following information:

```
SPCFORCES=ALL
DISP=ALL
GPFORCES=ALL
ELFORCES=ALL
SPC=100
```

(only for normal plate or solid rotations and the structure must be free at base)

```
SUBCASE 1
SUBTITLE=UNIT ENFORCED DISP IN X DIRECTION
LOAD=100
SUBCASE 2
SUBTITLE=UNIT ENFORCED DISP IN Y DIRECTION
LOAD=200
SUBCASE 3
SUBTITLE=UNIT ENFORCED DISP IN Z DIRECTION
LOAD=300
SUBCASE 4
SUBTITLE=UNIT ENFORCED ROTA ABOUT X AXIS
LOAD=400
SUBCASE 5
SUBTITLE=UNIT ENFORCED ROTA ABOUT Y AXIS
LOAD=500
```

```
SUBCASE 6
SUBTITLE=UNIT ENFORCED ROTA ABOUT Z AXIS
LOAD=600
```

BULK DATA must include the following information: (The displaced grid is **grid point 17104** in this example.)

```
BEGIN BULK
SPCD    100     17104   1       1.0
SPCD    200     17104   2       1.0
SPCD    300     17104   3       1.0
SPCD    400     17104   4       1.0
SPCD    500     17104   5       1.0
SPCD    600     17104   6       1.0

$ SPC FOR UNIT ENFORCED DISP CHECK
SPC1    100     123456  17104
FORCE   100     17104   0       0.0     1.0     0.0     0.0
FORCE   200     17104   0       0.0     0.0     1.0     0.0
FORCE   300     17104   0       0.0     0.0     0.0     1.0
MOMENT  400     17104   0       0.0     1.0     0.0     0.0
MOMENT  500     17104   0       0.0     0.0     1.0     0.0
MOMENT  600     17104   0       0.0     0.0     0.0     1.0
```

Unit Enforced Displacement or Rotation with Different Output Coordinate Systems

EXECUTIVE DECK must include the following information:

```
SOL 24

$ CAUSES GRID DISPACEMENTS TO BE OUTPUT IN THE BASIC COORD. SYSTEM
$ !WARNING! OUTPUT ONLY DISPLACEMENTS
$ *********FORCES AND SUCH MAY NOT BE CORRECT*********
COMPILE SOL24,SOUIN=MSCSOU
ALTER 160 $
PARAM //'ADD'/ALWAYS/-1/0 $
VECPLOT UGV,BGPDT,EQEXIN,CSTM,CASECC,/UGVBASIC//0/1 $
EQUIV UGVBASIC,UGV/ALWAYS $
ENDALTER
```

CASE CONTROL must include the following information:

```
$SPCFORCES=ALL
DISP=ALL
$GPFORCES=ALL
$ELFORCES=ALL
SPC=100   $(only for normal plate and solid rotations
$          and the structure must be free at base)
SUBCASE 1
SUBTITLE=UNIT ENFORCED DISP IN X DIRECTION
LOAD=100
SUBCASE 2
SUBTITLE=UNIT ENFORCED DISP IN Y DIRECTION
LOAD=200
SUBCASE 3
SUBTITLE=UNIT ENFORCED DISP IN Z DIRECTION
LOAD=300
SUBCASE 4
SUBTITLE=UNIT ENFORCED ROTA ABOUT X AXIS
LOAD=400
SUBCASE 5
SUBTITLE=UNIT ENFORCED ROTA ABOUT Y AXIS
LOAD=500
SUBCASE 6
SUBTITLE=UNIT ENFORCED ROTA ABOUT Z AXIS
LOAD=600
```

BULK DATA must include the following information:

```
BEGIN BULK
SPCD    100     17104   1       1.0
SPCD    200     17104   2       1.0
SPCD    300     17104   3       1.0
SPCD    400     17104   4       1.0
SPCD    500     17104   5       1.0
SPCD    600     17104   6       1.0
$ SPC FOR UNIT ENFORCED DISP CHECK
SPC1    100     123456  17104
```

```
MOMENT  400     17104   0       0.0     1.0     0.0     0.0
MOMENT  500     17104   0       0.0     0.0     1.0     0.0
MOMENT  600     17104   0       0.0     0.0     0.0     1.0
FORCE   100     17104   0       0.0     1.0     0.0     0.0
FORCE   200     17104   0       0.0     0.0     1.0     0.0
FORCE   300     17104   0       0.0     0.0     0.0     1.0
```

Free-Free Dynamics with a Stiffness Equilibrium Check

Purpose

The Free-Free Dynamics with a Stiffness Equilibrium check verifies that the model will act as a rigid body when it is unconstrained. It also checks the stiffness matrix to verify that it doesn't contain any grounding effects, such as illegally specified SPC or rigid elements. To perform this check, remove all external constraints (usually SPCs) and perform standard normal modes run recovering at least the first six modes.

Output Results

The model should obtain a minimum of six rigid body modes. These modes should have frequencies less than or equal to 1.0E-04 Hz. The GROUNDCHECK case control card should be used as a stiffness equilibrium. Strain energy filters are set by the THRESH parameter in the card. By default, the threshold is calculated by dividing the largest term in the stiffness matrix by 1.0e+10. The output will indicate whether the check is passed or failed.

The equilibrium check gives possible grounding in the model. In the output file look G-SET, N-SET, F-SET, and A-SET. The maximum strain energy for each DOF will be shown in these tables as well as whether or not it passes or fails. The following descriptions give possible problems from a particular table. Note that an error in a given set will cause errors to appear in subsequent sets, regardless of if any exist.

G-set

- CELAS problems. Non-coincident grids or misaligned output coordinate systems (most common).
- DMIG or DMAP inserted matrices associated with GRIDs. For the latter they should obviously be inserted before the check.
- Bad NASTRAN elements. This is rare but possible. Also small errors from a lot of elements will eventually add up, so very large models often have relatively poor equilibrium.

N-set

- Incorrect or imprecise MPC equations. Usually similar to CELAS problems. NASTRAN runs with improperly inserted Craig-Bampton models will show errors in this table if the boundary, as well as modes, were placed at SPOINTs. This is because this is the first set where geometry is associated with the boundary.

F-set

- Incorrect SPC's. Note that a constrained model will obviously show grounding.

A-set

- The model is constrained to ground (and not intended to be Free-Free).
- The reference GRID point in the GROUNDCHECK command is located too far from the model CG. The GRID parameter is optional.
- PARAM,AUTOSPC,YES constrains near-singular DOF.

Input Required

EXECUTIVE DECK must include the following information:

```
SOL 103 for MSC
```

CASE CONTROL must include the following information:

```
GROUNDCHECK(SET=ALL)=YES
$ (this will print out all the set info - sample output below)
SPCFORCES=ALL
DISP=ALL
SPC=100
$ (only for normal plate rotations, structure must free at base)
METHOD= 20
```

BULK DATA must include the following information:

```
EIGR, 20, $ (Use any method you want to obtain all rigid bodies modes and the
$ first flexible mode.  We recommend LANCZOS.)
```

Sample GROUNDCHECK Output

```
*** USER INFORMATION MESSAGE 7570 (GPWG1D)
     RESULTS OF RIGID BODY CHECKS OF MATRIX KGG      (G-SET)  FOLLOW:
     PRINT RESULTS IN ALL SIX DIRECTIONS AGAINST THE LIMIT OF   9.216514E+00
           DIRECTION        STRAIN ENERGY        PASS/FAIL
           ---------        -------------        ---------
             1               2.098354E-05          PASS
             2               6.123831E-06          PASS
             3               3.367564E-08          PASS
             4               1.326509E-03          PASS
             5               9.882150E-04          PASS
             6               4.239914E-04          PASS

     SOME POSSIBLE REASONS MAY LEAD TO THE FAILURE:
       1. CELASI ELEMENTS CONNECTING TO ONLY ONE GRID POINT;
       2. CELASI ELEMENTS CONNECTING TO NON-COINCIDENT POINTS;
       3. CELASI ELEMENTS CONNECTING TO NON-COLINEAR DOF;
       4. IMPROPERLY DEFINED DMIG MATRICES;
.
.
.
*** USER INFORMATION MESSAGE 7570 (GPWG1D)
     RESULTS OF RIGID BODY CHECKS OF MATRIX KNN      (N-SET)  FOLLOW:
     PRINT RESULTS IN ALL SIX DIRECTIONS AGAINST THE LIMIT OF   9.216514E+00
           DIRECTION        STRAIN ENERGY        PASS/FAIL
           ---------        -------------        ---------
             1               2.098167E-05          PASS
             2               6.120336E-06          PASS
             3               3.724166E-08          PASS
             4               1.326376E-03          PASS
             5               9.887038E-04          PASS
             6               4.241481E-04          PASS

     SOME POSSIBLE REASONS MAY LEAD TO THE FAILURE:
```

1. MULTIPOINT CONSTRAINT EQUATIONS WHICH DO NOT SATISFY RIGID-BODY MOTION;
2. RBE3 ELEMENTS FOR WHICH THE INDEPENDENT DEGREE-OF-FREEDOM CANNOT DESCRIBE ALL POSSIBLE RIGID-BODY MOTIONS.

.
.
.
*** USER INFORMATION MESSAGE 7570 (GPWG1D)
RESULTS OF RIGID BODY CHECKS OF MATRIX KFF (F-SET) FOLLOW:
PRINT RESULTS IN ALL SIX DIRECTIONS AGAINST THE LIMIT OF 9.216514E+00

DIRECTION	STRAIN ENERGY	PASS/FAIL
1	2.098167E-05	PASS
2	6.120336E-06	PASS
3	3.724166E-08	PASS
4	1.326376E-03	PASS
5	9.887038E-04	PASS
6	4.241481E-04	PASS

SOME POSSIBLE REASONS MAY LEAD TO THE FAILURE:
1. CONSTRAINTS WHICH PREVENT RIGID-BODY MOTION.
.
.
.
*** USER INFORMATION MESSAGE 7570 (GPWG1D)
RESULTS OF RIGID BODY CHECKS OF MATRIX KAA1 (A-SET) FOLLOW:
PRINT RESULTS IN ALL SIX DIRECTIONS AGAINST THE LIMIT OF 9.216514E+00

DIRECTION	STRAIN ENERGY	PASS/FAIL
1	2.098167E-05	PASS
2	6.120336E-06	PASS
3	3.724166E-08	PASS
4	1.326376E-03	PASS
5	9.887038E-04	PASS
6	4.241481E-04	PASS

IF THE MODEL HAS PASSED THE PREVIOUS CHECKS FOR THE G-SET AND N-SET,
THEN SOME POSSIBLE CAUSES ARE:
1. THE MODEL IS NOT INTENDED TO BE FREE-FREE WHICH INDICATES THAT THE MODEL IS PROPERLY CONSTRAINED TO GROUND;
2. THE REFERENCE GRID POINT (GRID=GID ON THE GROUNDCHECK COMMAND) IS LOCATED TOO FAR FROM THE MODEL'S CENTER OF GRAVITY. IT IS RECOMMENDED THAT THE REFERENCE GRID POINT BE LOCATED AS CLOSE AS POSSIBLE TO THE MODEL'S CENTER OF GRAVITY OF THE MODEL (SEE THE GRID POINT WEIGHT GENERATOR OUTPUT);
3. PARAM,AUTOSPC,YES CONSTRAINS NEAR-SINGULAR DEGREES-OF-FREEDOM. WHEN A FINITE ELEMENT MODEL WITH AUTOSPC FAILS THE A-SET CHECK, IT IS NOT EVIDENT THAT GROUNDING HAS OCCURRED. THE USE OF PARAM,SNORM WILL NOT ELIMINATE THE SPURIOUS FAILURE.

Unit Gravity Loading in All Three Translational Directions

Purpose

The Unit Gravity Loading validity check verifies that the model will provide accurate displacements and reactions forces under gravity loading. This is a good check to perform if your model will be used for quasi-static loads analysis.

Output Results

No large displacements should be found as a result of this validity check. In addition, forces and moments at only the mounting locations should be obtained when SPCFORCES=ALL is used in the CASE CONTROL. The forces should add up to the total weight of the model in the input loading direction and the forces in the other directions should be zero.

Input Required

EXECUTIVE DECK must include the following information:

```
SOL 24 or 101 for MSC
```

CASE CONTROL must include the following information:

```
SPCFORCES=ALL
DISP=ALL
SPC=100 (both normal plate rotations and structure held at base)
SUBCASE 1
 LABEL=UNIT GRAVITY LOADING IN X DIRECTION
 LOAD=139997
SUBCASE 2
 LABEL=UNIT GRAVITY LOADING IN Y DIRECTION
 LOAD=139998
SUBCASE 3
 LABEL=UNIT GRAVITY LOADING IN Z DIRECTION
 LOAD=139999
```

BULK DATA must include the following information:

```
PARAM   GRDPNT  0
GRAV    139997  0       386.4   1.      0.      0.
GRAV    139998  0       386.4   0.      1.      0.
GRAV    139999  0       386.4   0.      0.      1.
$ (386.4 in/sec2, use PARAM WTMASS .00259 in BULK DATA)
```

Unit Temperature Increase

Purpose

The Unit Temperature Increase validity check verifies that the model will provide accurate displacements due to temperature loading. This check should be performed on all models used for thermal distortion analyses.

Output Results

The displacements should be compared to the theoretical solution of alpha*l*deltaT. This comparison can be done easily when a sorted bulk data deck of the model is run through the program THDEF, located on "merger" in MERGER::SYS$SYSDISK:[USER,THDEF]. The program will provide a file containing the theoretical displacements for the model. The displacements from NASTRAN should be the same as the displacements from the THDEF program. The model must contain a kinematic mount to run the THDEF program. Also, the NASTRAN model should contain a kinematic mount for this validity check. A limitation to the THDEF program is that it does not handle local coordinate systems.

The grid point force balance should be zero for all grid points and only forces at the mounting locations should be output when SPCFORCES=ALL is included in the CASE CONTROL. Look at the displacements of the model in your post-processor to make sure it expands as expected.

Input Required

It is suggested you create a kinematic mount for your model, e.g., one GRID point constrained in all 6 DOFs.

EXECUTIVE DECK must include the following information:

```
SOL 24 or 101 for MSC
```

CASE CONTROL must include the following information:

```
SPCFORCES=ALL
DISP=ALL
GPFORCES=ALL
SPC=100 (both normal plate rotations and kinematically mounted structure)
TEMP=50
```

BULK DATA must include the following information:

```
TEMPD   50      21.  (reference temperature + 1.)
```

RBE2s that are not joined by coincident nodes should not be used.

Change all MAT1, MAT8, MAT9, etc., to contain the same thermal expansion coefficient and the same reference temperature (20. in this example).

Yield Stress Notes

A Few Definitions

Yield Stress = sigma y
Ultimate Stress = sigma u

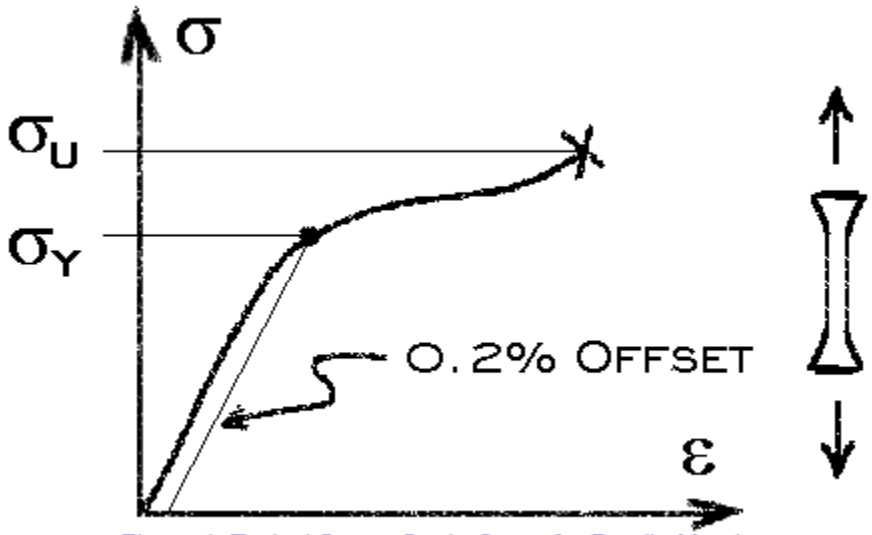


Figure 1. Typical Stress-Strain Curve for Ductile Metals

Sigma y and sigma u are determined from uniaxial tensile tests. Since the exact point at which yield occurs may not be visually obvious it must be defined in some other consistent manner. For most ductile metals the following convention has been widely accepted. The yield point is determined using a line parallel to the linear portion of the curve and originating at epsilon = 0.002 (The 0.2% offset method).

Since several tests of the same material will result in slightly different values for sigma y and sigma u, a statistical approach must be used to determine the exact value that will be used. For example: stress values listed in the military handbook (MIL-HDBK-5F) are catagorized as A-Basis and B-Basis. Quoting from the hand book:

- A-Basis. -- At least 99 percent of the population of values is expected to equal or exceed the A basis mechanical property allowable, with a confidence of 95 percent.
- B-Basis. -- At least 90 percent of the population of values is expected to equal or exceed the B basis mechanical property allowable, with a confidence of 95 percent.

From this it can be seen that just determining what value to use for the yield stress requires some thought.

A Few More Definitions (for bi-axial stress)

Normal Stress in x direction - sigma x
Normal Stress in y direction - sigma y (not the same as yield stress)
Shear Stress in xy plane - tau xy
Principal Stresses - sigma 1 and sigma 2

For a 2D plate the state of stress at a point can be described using any arbitrary axis. The state of stress using this axis will be described by the values sigma x, sigma y, and tau xy.

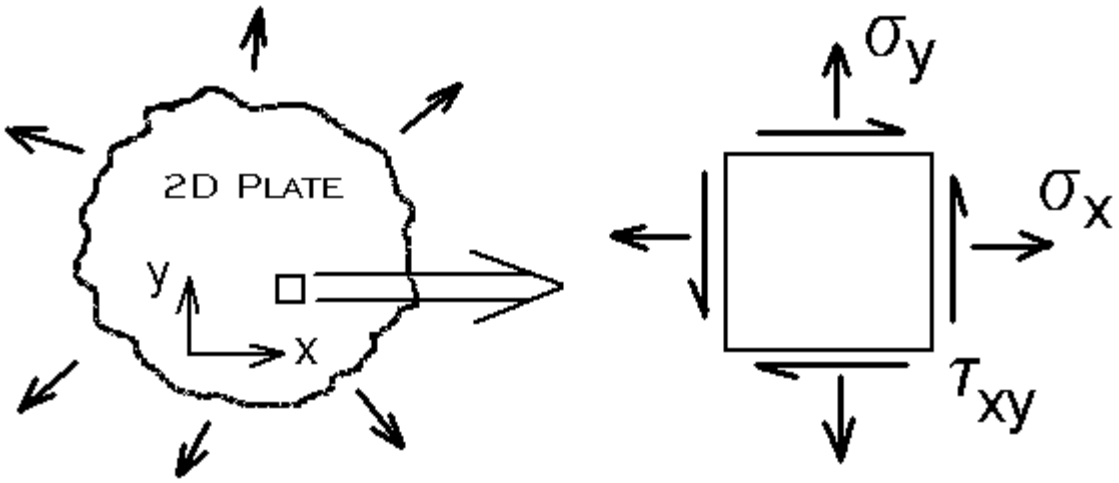


Figure 2. State of Stress Described Using Arbitrary Axis

There will exist some orientation of the axis at which when the state of stress is described, the shear stress will be zero. The axis when oriented at this angle is called the principal axis. The normal stresses for this special orientation are defined as the principal stresses, sigma 1 and sigma 2.

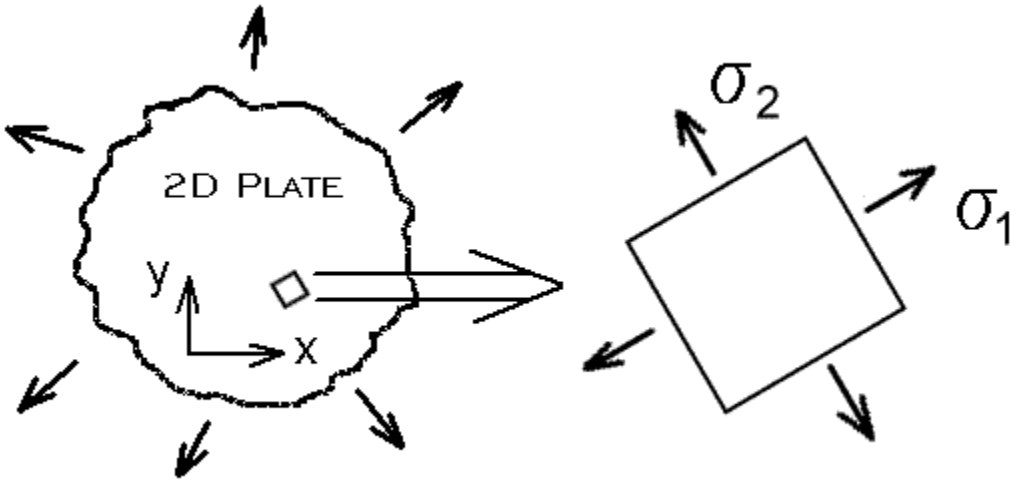


Figure 3. State of Stress Described Using the Principal Axis (tau xy = 0)

Tresca Yield Criteria
(Max Shear Stress Yield Criteria)

Tresca postulated that a material under a multi-axis state of stress will yield when the Max Shear Stress reaches some critical value. Using this assumption the yield envelope for a bi-axial state of stress is as shown in Figure 4. (Note: sigma y in Figure 4 is the yield stress)

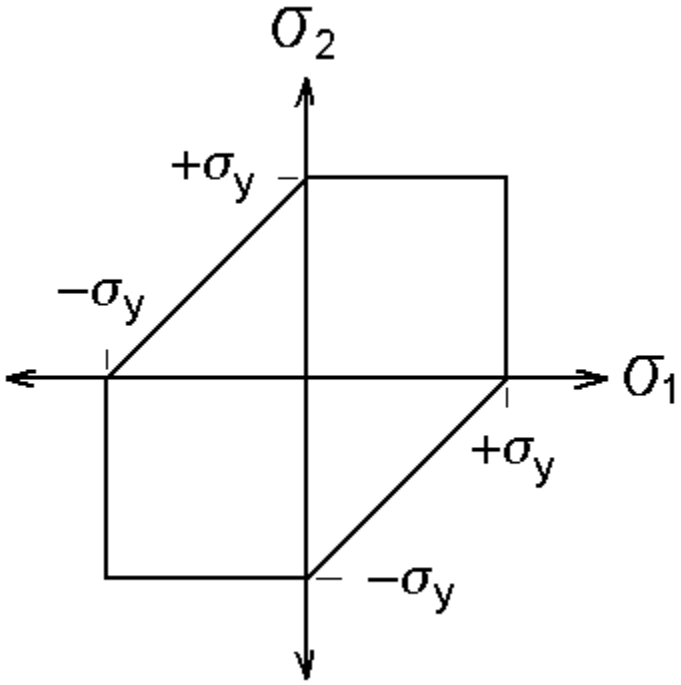


Figure 4. Tresca Yield Envelope

Why the yield envelope takes this shape can be illustrated by inspecting Morh's circle for various states of stress. First, consider the uniaxial tensile test at the moment yield occurs (Figure 5.). For this case one of the principal stresses will be equal to the yield stress and the other principal stress will be zero. This condition defines the critical value of the shear stress at which yield occurs. According to Tresca, yield will occur only if the shear stress exceeds this value. It should be noted that this value is half the value of the yield stress.

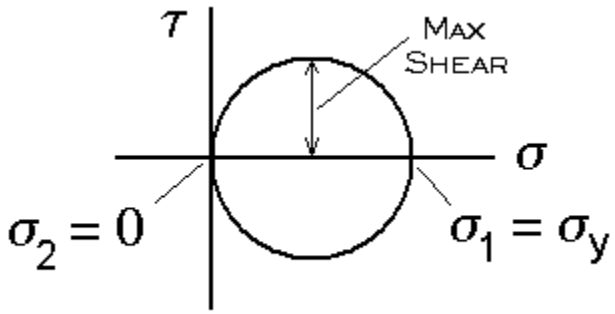


Figure 5. Mohr's Circle for Uniaxial Stress Test

If either of the principal stresses exceed the yield stress, the out of plane shear will exceed the Max Shear value (see Figure 6.). This is because for 2D problems the out of plane principal stress (sigma 3) is assumed to be zero. Therefore, neither sigma 1 nor sigma 2 may exceed sigma y.

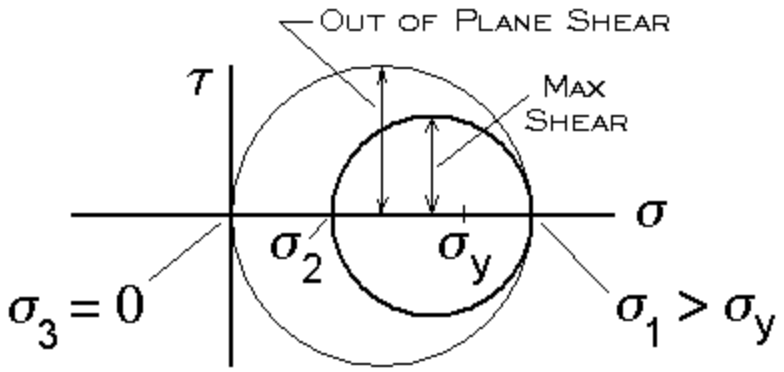


Figure 6. Mohr's Circle for sigma 1 or sigma 2 Greater than sigma y

In the case where one of the principal stresses has the opposite sign of the other (i.e. one in compression the other in tension. Quadrant 2 and 4 of the Tresca yield envelope) yield will occur before either of the principal stresses reaches the yield stress. As illustrated in Figure 7. if sigma 2 becomes more negative sigma 1 must also decrease (Other wise the shear stress will increase beyond the max value). Essentially you may slide a constant size circle along the axis between positive and negative sigma y.

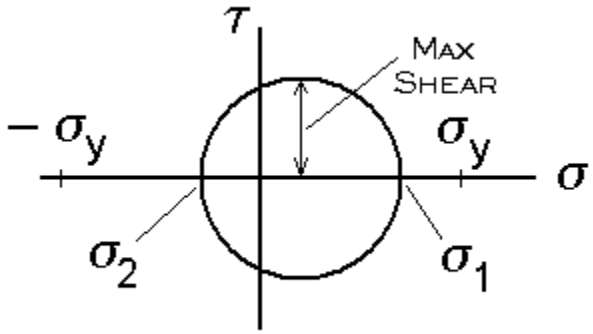


Figure 7. Mores Circle for Principal Stresses of Opposite Sign

Von Mises Yield Criteria

Von Mises postulated (1913) that a material will yield when the distortional energy at the point in question reaches a critical value. The distortional energy written in terms of the 2D principal stresses and the yield stress is as follows.

$$\left(\sigma_1 - \sigma_2\right)^2 + \sigma_1^2 + \sigma_2^2 = 2\sigma_y^2$$

Figure 8. shows the Von Mises Yield envelope superposed over the Tresca yield envelope. The scatter of test data is such that either criteria is as applicable as the other. For hand calculations Tresca is easy to implement; for a computer program Von Mises is convenient because the entire envelope can be represented with a single equation.

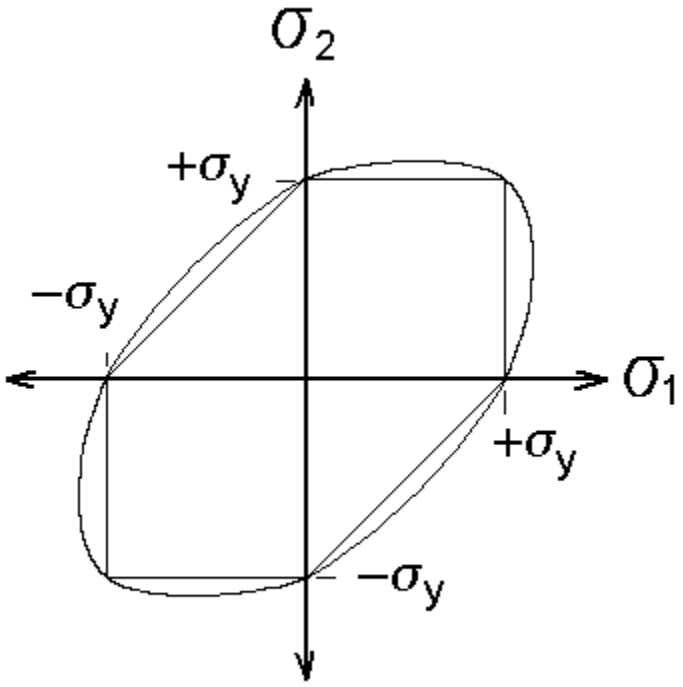


Figure 8. Von Mises Yield Envelope